

AMPHIPHILIC MARINE COATINGS SYSTEM OF SELF-LAYERED PDMS-PEG
SURFACE WITH AN EPOXY-POLYURETHANE MATRIX AND ITS CRITICAL
HYDROPHILIC-LIPOPHILIC RATIO (CHLR)

A Thesis
Submitted to the Graduate Faculty
of the
North Dakota State University
of Agriculture and Applied Science

By
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In Partial Fulfillment of the Requirements
for the Degree of
MASTER OF SCIENCE

Major Department:
Coatings and Polymeric Materials

March 2023

Fargo, North Dakota

North Dakota State University
Graduate School

Title

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State University's regulations and meets the accepted standards for the degree of

MASTER OF SCIENCE

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ABSTRACT

Settling of marine organisms on ships can cause biofouling, leading to undesired consequences. Fouling-release coatings are a promising approach to avoid biofouling. Today's commercial fouling-release coatings are mainly based on silicone elastomers, which are not ideal for slime release, and the coating is very soft. Robust coatings with excellent biofouling release performance are needed. To achieve such a goal, self-stratified, amphiphilic, robust glycidyl-carbamate (GC)-based coatings with good fouling-release properties were explored. Several factors influence coatings performance, including the molecular weight of poly (ethylene glycol) (PEG) and poly (dimethyl siloxane) PDMS moieties, the ratio of hydrophobic (PDMS) and hydrophilic (PEG) components in the system, and the effect of different amine curing agent.

ACKNOWLEDGMENTS

I am very grateful for everything I have experienced in the past few years, escape and encounter, like a dream, but extremely sincere.

First of all, I would like to express my sincere gratitude to my advisor, Dr. Dean Wester, for his continued support and guidance throughout the course of my work. I appreciate your trust, help, and mentorship. It may have been decided in the summer of 2007 that I would be your student when I bought a book called <Organic Coatings: science and technology>. Many thanks to my committee members, Dr. Mohiuddin Quadir and Dr. Yechun Wang. Thank you for your time and guidance.

I would also like to thank Dr. Chunju Gu and Greg Strommen for their training and support in instrument use. Thanks to Dr. Shane Stafslie, Lyndsi Vanderwal, and Jim Bahr for their collaborations in assessing my coatings' fouling-release performance and surface content. Thanks to Dr. David Boucher for his training at AFM.

Additionally, thanks to my groupmates and classmates, Dr. Jingbo Wu, Joseph Dahlgren, Michael Odegaard, Marta Vonsul, and Dr. Jackson Benda. I have benefited a lot from your optimism and sincerity.

Thank you to my parents and kids. Without their love and support, none of this work would be possible. Finally, to my wife, Qiuxiang Pan, I am forever grateful for your endless support, encouragement, companionship, comfort, and tolerance. Now we have one more relationship—alumni. I look forward to the rest of our lives together.

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CHAPTER 1. INTRODUCTION

1.1. Biofouling Introduction

Marine biofouling is known as the undesirable accumulation of microorganisms, algae, and animals on artificial surfaces immersed in seawater.¹ Due to the diversity of the ocean ecosystem, biofouling formation is particularly complex and poses severe challenges to human infrastructure, particularly shipping.

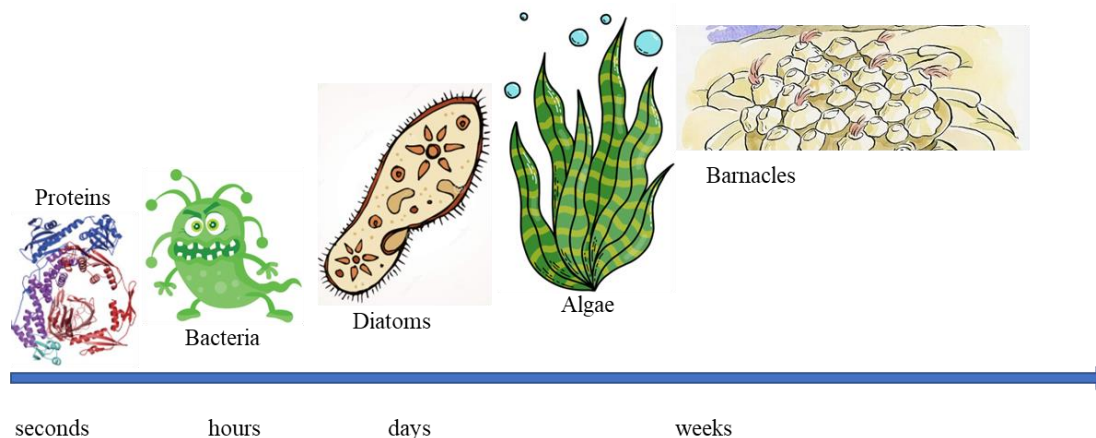
1.1.1. The complexity of marine life and marine organisms

Water has been described as the solvent of life due to its ability to dissolve many substances.² Covering 70% of the Earth's surface, the oceans provide about 80% of the living space of all life on Earth. More than 200,000 documented marine species and perhaps two million undocumented marine species, including the smallest and largest life, live in the oceans.³ The size of marine species ranges from smaller than 0.05 micrometers to more significant than 30 meters.⁴ Many marine microorganisms, like protists and bacteria, live in the oceans and are estimated to constitute about 70% or more of the total marine life. The temperature, salinity, pressure, light, currents, and other factors make the oceans very different, which makes the oceans have higher diversity.⁵ This diversity provides excellent help to humans, providing inexhaustible resources such as food. But it also brings significant challenges to the development of marine civilization, such as the extreme weather brought by the ocean and the biofouling that limits the performance and even damages artificial materials immersed or near seawater.

1.1.2. Biofouling formation process

It is estimated that more than 4,000 marine organisms can settle on the surface to form biofouling. Due to the diversity of the ocean, together with the variation of location,

environmental conditions, seasonal change, and the different attachment mechanisms involved, biofouling also has higher diversity.⁶ Biofouling formation cannot be explained simply by a single (universal) model. Nonetheless, there are many reports about the process and timeline of the biofouling process.⁷⁻⁹ An acceptable process can be described below according to Delauney et al.¹⁰ and as shown in Scheme 1.1.



Scheme 1.1. Biofouling formation process: The biofouling formation evolves with time. The biofouling phases may not appear one by one (can occur in parallel) and differ for different marine species. Reproduced from reference 6.

Almost immediately after immersion, a conditioning film will be first formed by the adsorption of organic and inorganic macromolecules (primarily proteins). Microbial cells will then be transferred to the surface in a few hours, absorbing bacteria and diatom, forming a microbial film on the surface. A few days later, a more complex community will be developed, and multicellular species, microalgae, debris, sediments, etc., will appear on the surface. Finally, the surface will be attached to large marine organisms such as barnacles, mussels, macroalgae, etc. The biofouling phases may not appear one by one (can occur in parallel) and differ for different marine species.

1.1.3. Impact and challenges of biofouling

Marine biofouling affects appearance and significantly impacts the economy, environment, and ecology in human production practice.

1.1.3.1. Economic impact

Marine biofouling has brought huge economic losses to human beings. For example, nowadays, more than 80% of global trade by volume is carried out by shipping.¹¹ Marine biofouling increases the thickness and roughness of the ship's surface, thereby increasing the friction force when the ship is running, increasing energy consumption, and causing huge financial losses. For the DDG-51 class of naval vessels (large size with a length of about 150 m), the overall expenses because of hull biofouling were estimated to be approximately \$56 M per year or \$1.0 B over 15 years. Even for the mid-sized naval ship, a high degree of fouling may cause an increase of 20.4% in fuel consumption. Not only that, the estimated cost for transport delays, hull repair, cleaning, and general maintenance caused by biofouling is 150 billion USD per year.^{12,13}

1.1.3.2. Environment and ecology impact

In addition to severe economic losses to human beings, biofouling also brings serious environmental and ecological issues. Firstly, the significant increase in fuel consumption will inevitably lead to a rise in emissions of carbon, chemical waste, and other harmful compounds. Cleaning, repairs, and abrasion of the ship hull and its protective coatings may generate substances that are very harmful to the marine environment.¹⁴ In addition, the organisms attached to the biofouling carried to far-away oceans by the widespread network of global shipping routes may pose a huge environmental threat to the local ocean ecosystem.¹⁵

1.2. Biofouling Prevention Coating Technologies

Since biofouling causes problems for human beings, human beings have wanted to use paint to prevent biofouling. After hundreds of years, many technologies have been developed. The general trend is to develop from toxic systems to environmentally friendly systems.

1.2.1. Toxic antifouling coatings

Coating technology has been applied to wooden ships and vessels and has been recorded in ancient times. Waxes, tar, asphalt, copper, lead, and other materials were used to protect the wooden hull, but at the same time, it also prevented biofouling until nearly the end of the 17th century. As iron components replaced wood as the primary material of the hull, the coating technology also changed. Especially in the 1940s, with the invention of artificial resins, more coating techniques were tried and used by humans. So-called antifouling coatings containing biocides were gradually being widely used to avoid biofouling. In the 1960s-1980s, tributyl-tin (TBT) was introduced into the antifouling coating system and widely used because of its extensive and effective bactericidal performance against biofouling. TBT-based antifouling coatings systems were restricted in most European countries and finally banned worldwide in 2008 because of the significant environmental concerns caused by the tributyl-tin (TBT) released into the sea by antifouling coatings.^{16,17} Low or non-toxic antifouling coating systems are increasingly being researched extensively.

1.2.2. Biocide-free fouling avoids coating systems

With new and strict environmental and health restrictions imposed on materials for marine applications, it became imperative to study biocide-free coating technologies and strategies for avoiding biofouling. Two main methods were used to prevent the formation of biofouling: ‘the prevention of attachment of biofoulants and ‘detachment of biofoulants’.¹⁸ Many

factors, such as surface wettability, roughness, modulus, shape, etc., will affect the biofoulings' mechanical stability and adhesion to the substrate. The Baier curve (Fig. 1.1) showed the relationship between surface free energy and relative bio-adhesion and has been used to design biocide-free antifouling and fouling-release coatings.¹⁹

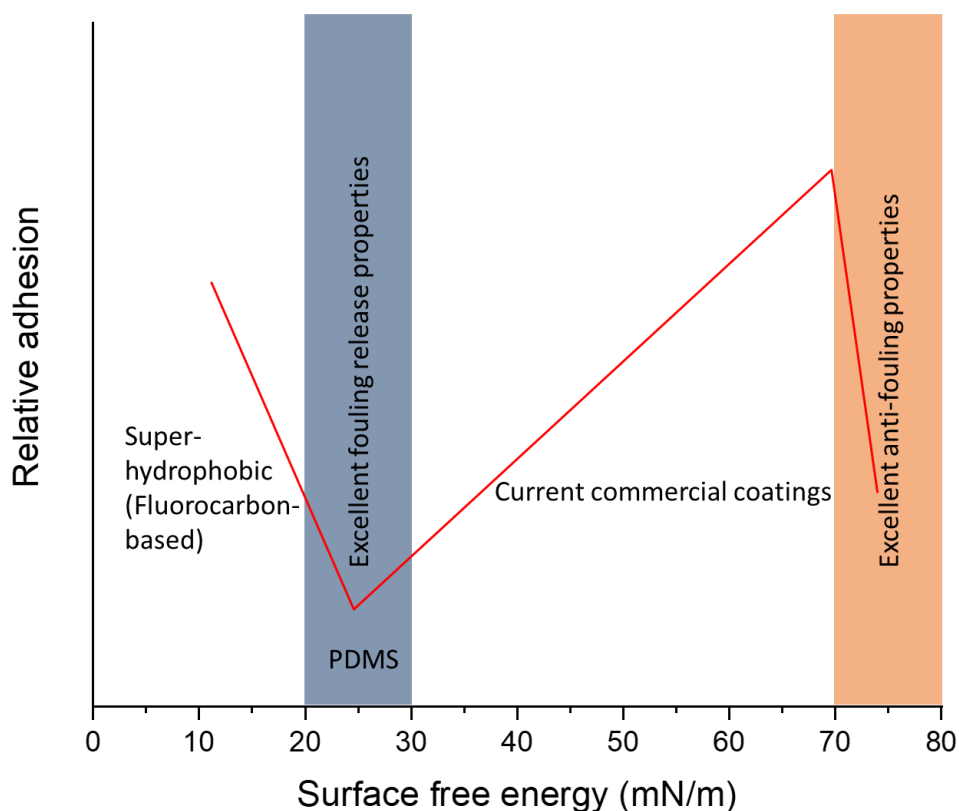


Figure. 1.1. Different surface free energies promote different fouling control effects. Qualitative representation of the Baier curve adapted from reference 19 showing the influence of the free energy of the coating surface on the relative adhesion of a bio-organism on that coating.

A surface with free energy above 70 mN/m performs well as an antifouling coating due to the formed hydration layer, which can avoid the adhesion of proteins, thereby preventing the generation of conditioning film. In comparison, a surface with free energy between 20 and 30

mN/m has optimal fouling-release performance due to the reduced interactions between the organism and the substrate and thus exhibits minimum bio-adhesion.

1.2.2.1. Biocide-free antifouling coating

The ‘Prevention of attachment of biofoulants’ strategy is used to design antifouling coatings, as Fig.1.1 shows. One of this strategy's most critical points is to avoid forming a conditioning film (to prevent the adhesion of proteins) which will trigger the settlement process. Besides using toxic additives to prevent biofouling, researchers also found that (super-) hydrophilic surfaces have good antifouling properties due to the strong surface hydration layer, which acts as a physical and energy barrier for the marine organism's settlement. Hydrophilic polymers include poly (ethylene glycol)-based, acrylate-based and zwitterionic polymers which can be fabricated as hydrogels or surface-grafted brushes with excellent antifouling properties were widely studied.²⁰⁻²² PEG is one of the most successful hydrophilic polymers applied in antifouling coatings in recent studies.^{23,24} The low PEG-water interfacial energy, the hydration layer is formed due to the hydrogen bonding between the water molecules and the ether oxygen atoms, the mobility and flexibility of the PEG chains, PEG's electrical neutrality, and steric hindrance are the main reasons why PEG-based coatings have excellent antifouling properties.

1.2.2.2. Biocide-free fouling release coating

The fouling release coating system is designed according to the ‘detachment of biofoulants’ strategy. The principle of this strategy is to create a weak connection between the surface and the biofoulants so that it will easily fail due to the stress caused by the weight of fouling or the hydrodynamic flow generated by the ship's navigation.²⁵⁻²⁷ Many materials with low surface energy, such as silicone-based materials, fluorine-based materials, and combined fluorine–silicone-based materials, were have been used to prepare fouling release coating.^{25,28-31}

Silicone-based materials poly (dimethylsiloxane) (PDMS) based coatings are primarily utilized for fouling release applications mainly due to their low critical surface energy, nearly absent micro-roughness, and low modulus. Because of a particular structure and the partially ionic (non-directional) nature of the Si–O bonds,²⁷ PDMS surfaces show conformational mobility, which allows close packing of the pendant methyl groups at the interface but are also highly mobile, which reduces the adhesion of marine organisms.³²⁻³⁵ It is worth noting that oceanic slime layers composed of bacteria, diatoms, and a matrix of secreted extracellular polymeric substances are not readily detached from PDMS-based fouling release coatings, even at high operating speeds. In addition, poor adhesion to substrates and low resistance to mechanical damage limit their application in the marine sector.^{36,37}

1.2.2.3. Amphiphilic marine coating

An amphiphilic surface possessing both hydrophilic (water-loving, polar) and lipophilic (fat-loving, non-polar) properties,⁶ could be created by combining the above-mentioned hydrophobic and hydrophilic materials. The amphiphilic surface exhibits promising fouling control properties and durability.³⁸⁻⁴¹ An amphiphilic marine coating often has excellent antifouling and fouling release properties simultaneously. A generally accepted hypothesis is that the amphiphilic surface ‘confuses’ the biofoulants: amphiphilic materials provide a heterogeneous nanoscale mosaic chemical surface, where the coexistence of hydrophobic and hydrophilic domains can confuse organisms during settlement and adhesion. Whether hydrophobic or hydrophilic, marine organisms cannot be stably adsorbed on the surface due to the surface’s amphiphilicity, which causes a good fouling release function of the amphiphilic surface. Amphiphilic coatings can be prepared by using amphiphilic copolymers as basic materials, incorporating amphiphilic copolymers as surface-active additives, and grafting the

amphiphilic components to the coating surfaces. PDMS is used most to provide hydrophobicity, while PEG is used to provide hydrophilicity. Amphiphilic copolymers combined with PDMS and PEG moieties can be synthesized through various approaches, such as sol-gel chemistry, free radical polymerization, and atom-transfer radical polymerization.⁴²⁻⁴⁴ Amphiphilic surfaces with PDMS and PEG domains can offer a broad spectrum of efficacy and are currently conceived as one more promising strategy for marine biofouling control applications in the last decade.

1.3. Research Objectives

This research has two primary goals:

- (1) Continuing Dr. Rahimi's prior research⁴⁵ in Chapter 2, preparing novel amphiphilic fouling release coatings based on glycidyl-carbamate (GC) system, using PDMS and PEG with different molecular weights as amphiphilic domains, studying the effects of different curing agents.
- (2) Studying the effects of the molar ratio of PDMS to PEG in Chapter 3, trying to discover the critical molar ratio of PDMS and PEG, at which fouling release coatings have good fouling control balance.

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CHAPTER 2. AMPHIPHILIC MARINE COATING SYSTEMS OF SELF-STRATIFIED PDMS-PEG SURFACES WITH AN EPOXY-POLYURETHANE MATRIX CURED WITH DIFFERENT CURING AGENTS

2.1. Introduction

The diversity of marine organisms brings enormous challenges to human exploration and interaction with the ocean. For example, some marine organisms will adhere and settle on the surface of ships, forming marine biofouling, which will increase the drag of ships and then increase fuel consumption and transport non-indigenous species. For example, it was reported that the US Navy spends much money to deal with the issues caused by marine biofouling.¹ Because of the diversity of marine organisms, biofouling is a complex process that is non-linear and involves the formation of tiny bacteria that form a biofilm, which can occur just seconds after immersion, or larger macro foulants like barnacles and tubeworms that develop later. It is not easy to find a way to keep all marine organisms away from it. Humans have tried many methods to avoid or slow down the generation of marine biofouling on the surface of ships.²⁻⁵ The most direct way is to kill all marine organisms which try to contact the surface, and many marine coatings containing biocides have been developed and commercialized. However, biocide-containing coatings bring another severe problem once the biocides are released into the ocean. They not just only kill marine organisms trying to touch the surface of the ship but also other marine organisms, causing a severe ecological crisis.^{6,7} Therefore, marine coatings without biocides are urgently needed. Much research has taken place in this field. The most successful commercial products are the biofouling-release marine coatings prepared from silicone elastomers. Low modulus and low surface energy help silicone elastomer-based FR coatings reduce marine organisms' adhesion. However, due to the diversity of marine coatings, some

marine organisms are more likely to adhere to hydrophobic surfaces, which is why silicone elastomer biofouling release coatings have a poor effect on various slimes. In addition, due to the low hardness of silicone elastomers, the paint film is easily damaged, which requires frequent repairs.¹

The amphiphilic coating system is believed to have a good release performance for both hydrophilic and hydrophobic marine organisms. In previous work in our team, robust hydrophobic siloxane-polyurethane (SiPU) FR coatings with good mechanical durability were developed. Poly(ethylene glycol) (PEG)-modified amphiphilic SiPU systems have been studied and shown desirable FR performance.⁸⁻¹⁴ But this system still contains poly-isocyanate, which is not good for the workers' health due to its toxicity. Work in our group showed that glycidyl-carbamate (GC) resins, which do not contain poly-isocyanate but has urethane linkages, with crosslinking through epoxy chemistry, has excellent mechanical properties.¹⁵⁻¹⁷ Rahimi combined the GC resin coating system and amphiphilic coating system to develop an amphiphilic glycidyl-carbamate (GC) coating system. Poly(dimethyl siloxane)(PDMS) and PEG with a fixed weight ratio at 1:1 were used to synthesize GC prepolymers, providing amphiphilic features in the coating system. Ancamide® 702B75 and Amicure® PACM were used as curing agents to investigate the effect of curing agents, and the total amount of PEG and PDMS in the system (5% or 10%) was optimized.¹⁸ The objective of the work in this study was to continue Alireza Rahimi's work and examine the impact of more than eight curing agents, including polyetheramine, dimer diamine, and cycloaliphatic amine. The same method was used to synthesize GC resin and GC prepolymer and used a wider variety of curing agents to investigate the effect of curing agents. The selected curing agents, including the main commercial amine curing agents such as polyetheramine, dimer diamine, and cycloaliphatic amine, a total of 8

curing agents were investigated. The system's PDMS and PEG weight ratio factor was changed to 10% and 15%. A total of 64 coating systems were prepared and evaluated in this study.

2.2. Experimental

2.2.1. Materials

Isophorone diisocyanate (IPDI) polyisocyanate trimer Desmodur Z4470 BA was provided by Covestro LLC. Monocarbinol-terminated polydimethylsiloxanes (CT-PDMS) with molecular weights of 5000 $\bar{M}_n$ (MCR-C18) and 10,000 $\bar{M}_n$ (MCR-C22) were purchased from Gelest, Inc. Methyl ethyl ketone (MEK), methyl amyl ketone (MAK), hydrobromic acid (5.7 M), standardized hydrochloric solution (0.1 N) was purchased from Alfa Aesar, Glycidol, poly(ethylene glycol) methyl ether (550 $\bar{M}_n$ and 750 $\bar{M}_n$), and dibutyltin diacetate (DBTDAc) were purchased from Sigma Aldrich. Toluene and glacial acetic acid were purchased from VWR. Jeffamine D230 and Jeffamine D400 were provided by Huntsman corporation, Priamine-1071, Priamine-1074, and Priamine_1075 were supplied by Croda Inc, EPIKURE™ 3300 (IPDA) and EPIKURE™ 3233 were provided by Hexion Inc, Mitsubishi Gas Chemical Inc provided 1,3-BAC (1,3-bis(aminomethyl)cyclohexane). Dow Corning provided silicone elastomer Silastic® T2 (T2), and AkzoNobel International Paint provided Intersleek® 700 (IS-700), Intersleek® 900 (IS-900), and Intersleek® 1100SR (IS-1100) as commercial fouling release coatings controls. Intergard 264 was also provided by International Paint and was used to prime aluminum panels (4" x 8" in., 0.6 mm thick, type A, alloy 3003 H14) purchased from Q-lab. Q-lab also provided steel panels (3" x 6" in., 0.2 mm thick, smooth finish) for mechanical tests.

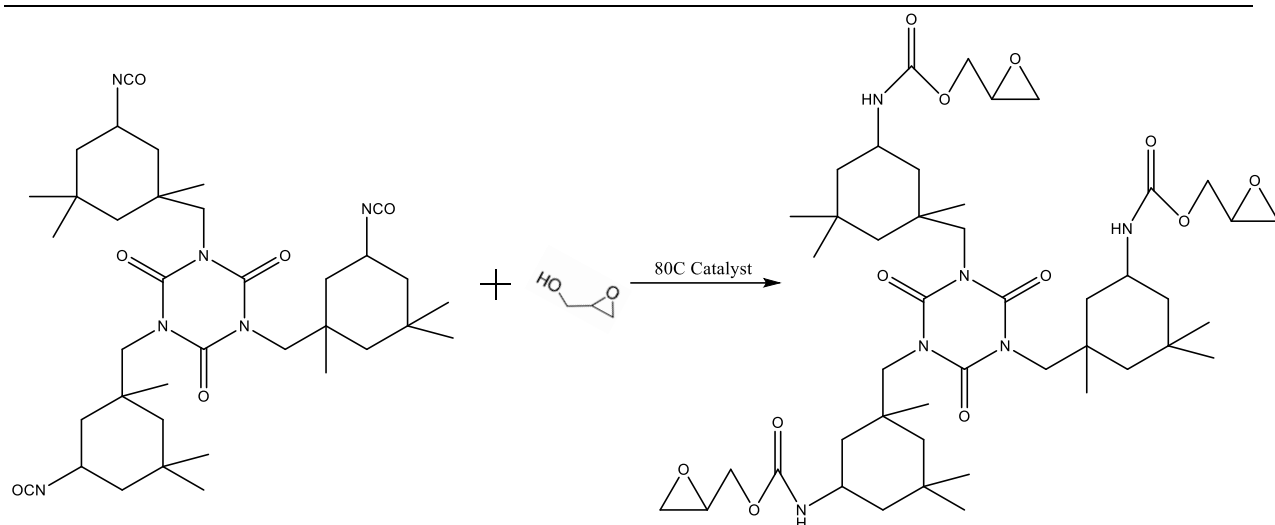
2.2.2. Synthesis of glycidyl carbamate (GC) resin

Glycidyl-carbamate (GC) resin,¹⁶ which is an epoxidized urethane-linkage-containing resin, was synthesized by the reaction between polyisocyanate IPDI trimer (Desmodur Z4470) and

glycidol as Scheme 1 shows, the ratio of -NCO: -OH was 1:1. As an example, 917.94 g of IPDI trimer (NCO eq wt. = 249.97), 197.23g glycidol, 55.76g DBTDAc catalyst solution (1% by wt. in MAK) and 359.83g toluene were added together in a 2L four-neck flask, equipped with a stirrer, nitrogen inlet, a reflux condenser, and temperature controller. The reaction was carried out at 80 °C for 2-3 hours.

Table 2.1. Compositions of GC resin and GC prepolymers

ID	PDMS MW ($\bar{M}_n$)	PEG MW ($\bar{M}_n$)	Wt.% of PDMS (PEG) in solid
GC resin	-	-	-
Prepolymer 5K550	5K	550	35.3
Prepolymer 5K750	5K	750	37.98
Prepolymer 10K550	10K	550	35.82
Prepolymer 10K750	10K	750	38.18

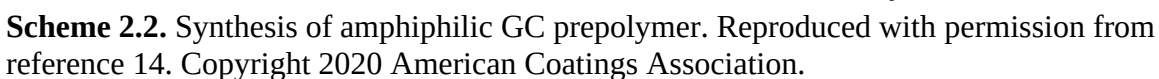


Scheme 2.1. Synthesis of glycidyl carbamate (GC) resin. Reproduced with permission from reference 14. Copyright 2020 American Coatings Association.

2.2.3 Synthesis of amphiphilic glycidyl carbamate prepolymers

Amphiphilic GC prepolymers were synthesized by the reaction between polyisocyanate IPDI trimer (Desmodur Z4470) and glycidol, PEG, and PDMS, as Scheme 2 shows. The ratio of NCO groups to the total OH groups was 1:1. Glycidol provided 33.3% of the OH groups, and the remaining 66.6% of OH groups were provided by PEG and PDMS, which were added in equal

weight ratios. As an example, to synthesize GC pre-polymer 10K PDMS-750 PEG, 100.00g of PEG 750 $\bar{M}_n$ (OH eq wt. = 750), 100.00g of Carbinol-PDMS 10K $\bar{M}_n$ (OH eq wt. = 10,000) were first diluted in toluene (135.57 g) in a 500-mL four-neck flask and mixed robustly with vortexing for two minutes. Glycidol (5.524 g) was added and mixed for another two minutes. Following this, 77.42g of IPDI trimer resin (NCO eq wt. = 262.53) and 13.60g of DBTDAC catalyst solution (1% by wt. in MAK) were added to the flask. The reaction was carried out at 80 °C for 2-3 hours. The flask was equipped with a stirrer, nitrogen inlet, reflux condenser, and temperature controller. In theory, the synthesized prepolymer contained 38.18 wt.% PEG and 38.18 wt.% PDMS. In this work, the molecular weight of PDMS and PEG was a key factor to be investigated, so two different molecular weights were used for PDMS (5000 $\bar{M}_n$ and 10,000 $\bar{M}_n$) and PEG (750 $\bar{M}_n$ and 550 $\bar{M}_n$), respectively. A total of four GC prepolymers were synthesized as Table 2.1 shows.



The total non-volatile solid content of GC resin and GC polymers was determined based on the guidelines outlined in ASTM 2369. An empty sample pan was weighted on an analytical balance to the nearest 0.1 mg, and this is the Pan Weight. To the tared pan was added ~1.0 g of sample and weight as was the initial weight—removed sample from balance. About 3 ml of solvent was added to cover the sample. Three samples were prepared and placed in the oven for 1 hour at 120 °C. Samples were removed from the oven, allowed to cool to room temperature and weighed as the Final Weight. The percent solids were calculated using equation (1). Results from the three samples should agree within ± 0.5 percentage points. If one sample is outside this limit, average

the remaining two results. If all three differed by more than this amount, the experiment was repeated.

$$\%Solids = \frac{FinalWeight - PanWeight}{InitialWeight} \times 100 \quad (1)$$

2.2.5. Epoxy equivalent weight titration

The epoxy equivalent weight (EEW) values of GC resin and GC prepolymers were determined using titration according to ASTM D1652. 1-2 g sample of GC resin or GC prepolymer was weighed in an Erlenmeyer flask, then dissolved in 15 mL of chloroform at room temperature. Crystal violet solution of 1% in glacial acetic acid was added (4-6 drops) as an indicator. The final solution in the flask was then titrated with a hydrobromic acid solution in glacial acetic acid until the blue solution changed to blue-green as the endpoint of the titration. A blank titration was also carried out using 15 mL pure chloroform without a sample. Three replicates were recorded for each GC resin/GC prepolymer. EEW was calculated according to equation (2) using the recorded information and the percent solids content.

$$EEW = \frac{1000 \times W}{N \times (V - B)} \quad (2)$$

where W = sample mass, in grams, N = normality of HBr solution, V = HBr solution used for titration, in mL, B = HBr solution used for blank, in mL.

The hydrobromic acid solution is standardized by titrating a 0.4g dried potassium acid phthalate (after drying one h @110 °C in the oven) solution in HOAc, following the same procedure as EEW titration, N was calculated according to equation (3).

$$N = \frac{1000 \times W}{204.2 \times V} \quad (3)$$

where W = KHP weight, in grams, V= HBr solution, in ml.

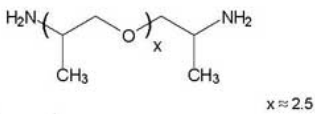
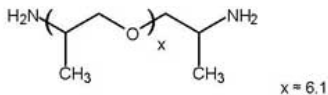
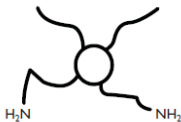
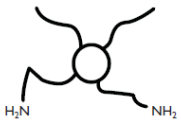
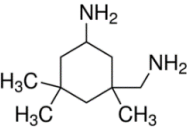
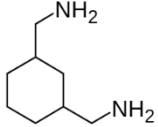
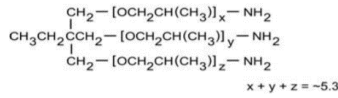
2.2.6. Fourier transform infrared spectroscopy

A Thermo Scientific Nicolet 8700 FTIR was used to run Fourier transform infrared (FTIR) spectroscopy of GC resin and GC prepolymers. A blank potassium bromide (KBr) plate was scanned first to get the background spectrum. The GC resin and GC prepolymers solution in acetone was spread as a thin layer on the same potassium bromide (KBr) plate to collect the spectrum with the background subtracted.

2.2.7. Coating formulation design and curing

In this work, eight different curing agents, which include the type of polyetheramine, dimer diamine, and cycloaliphatic amine, were used to study the effect of curing agents on the performance of coatings. Jeffamine D230 and Jeffamine D400 are polyetheramines and have the same structure but different chain lengths and molecular weights. Priamine_1071, Priamine 1074, and Priamine_1071 are dimer diamines with high hydrophobicity. Isophoronediamine (IPDA) and 1,3-bis(aminomethyl)diamine (1,3-BAC) are cycloaliphatic amines with different steric hindrances. EPIKURE™ 3233 is a trifunctional polyetheramine with high hydrophilicity, Table 2.2 shows the details of curing agents and their structures.

Table 2.2. Properties and structures of curing agents used in this study.

Curing Agent (Trade name)	Type	Amine H Eq. Weight (AHEW), g/eq	Structure
Jeffamine D230	Polyetheramine	60	
Jeffamine D400	Polyetheramine	115	
Priamine_1071	Dimer diamine	140	
Priamine_1074	Dimer diamine	135	 Large hydrocarbon part(C36)
Priamine_1075	Dimer diamine	140	
EPIKURE™ 3300 (IPDA)	Cycloaliphatic amine	43	
1,3-BAC	Cycloaliphatic amine	35.6	
EPIKURE™ 3233	Polyetheramine	81	

For each curing agent, four factors were examined: the molecular weights of PEG and PDMS and two levels (10 wt.% and 15 wt.% based on solid contents.) of the amounts of each of them used in the coating system. A full factorial experimental design was followed to assess these factors, involving eight coating formulations for each curing agent. A total of 64 formulations were prepared and labeled from C1 to C64, as Table 2.3 shows. Coatings are

labeled based on the content of PDMS, and the type of prepolymers, which are named based on the MW of their polymeric blocks, where the first is always MW of PDMS chain(s) and the second number is MW of PEG.

For each formulation, the GC resin and amphiphilic GC prepolymer were mixed in a vial and stirred for 24 hours. The curing agent (i.e., Jeffamine D230) was then added to the vial and stirred for another 20 minutes. The formulation was allowed to stand for 15 minutes to ‘sweat-in’ before application. The formulation was applied to 8” x 4” aluminum substrates using a drawdown bar with a wet film thickness of 80 μm . Coatings were cured ambiently for 24 hours and following oven-cured at 80 °C for 45 minutes. Formulations for the resin and prepolymers described earlier are provided in Table 1. To formulate coating, 10% 10K750 (C1), for example, 28.67 g GC resin (69.75% solids), 14.41g amphiphilic GC prepolymer (59.81% solids), and 3.92 g Jeffamine D230 curing agent were used. All outlined formulations for this study were prepared in the same way.

Table 2.3. Formulations of the coatings used in this study.

Formulation	Formulation Type	PDMS MW ($\bar{M}_n$)	PDMS Wt. %	PEG MW ($\bar{M}_n$)	PEG Wt. %	Curing agent
C1	10% 10K750	10,000	10	550	5	Jeffamine D230
C2	10%10K550	10,000	10	750	5	
C3	10% 5K750	5,000	10	550	5	
C4	10% 5K550	5,000	10	750	5	
C5	15% 10K750	10,000	15	550	10	
C6	15%10K550	10,000	15	750	10	
C7	15% 5K750	5,000	15	550	10	
C8	15% 5K550	5,000	15	750	10	
C9	10% 10K750	10,000	10	550	5	
C10	10% 10k550	10,000	10	750	5	
C11	10% 5K750	5,000	10	550	5	
C12	10%5K550	5,000	10	750	5	

Table 2.3. Formulations of the coatings used in this study (continued).

C13	15% 10K750	10,000	15	550	10	Jeffamine D400
C14	15%10K550	10,000	15	750	10	
C15	15% 5K750	5,000	15	550	10	
C16	15% 5K550	5,000	15	750	10	
C17	10% 5K550	5,000	10	550	5	Priamine_1071
C18	10%5K750	5,000	10	750	5	
C19	10% 10K750	10,000	10	550	5	
C20	10% 10K550	10,000	10	750	5	
C21	15% 10K750	10,000	15	550	10	
C22	15%10K550	10,000	15	750	10	
C23	15% 5K750	5,000	15	550	10	
C24	15% 5K550	5,000	15	750	10	
C25	10% 10K750	10,000	10	550	5	Priamine_1074
C26	10% 10K550	10,000	10	750	5	
C27	10% 5K750	5,000	10	550	5	
C28	10%5K550	5,000	15	750	5	
C29	15% 10K750	10,000	15	550	10	
C30	15%10K550	10,000	15	750	10	
C31	15% 5K750	5,000	15	550	10	
C32	15% 5K550	5,000	15	750	10	
C33	10% 5K550	5,000	10	550	5	Priamine_1075
C34	10%5K750	5,000	10	750	5	
C35	10% 10K750	10,000	10	550	5	
C36	10% 10K550	10,000	10	750	5	
C37	15% 10K750	10,000	15	550	10	
C38	15%10K550	10,000	15	750	10	
C39	15% 5K750	5,000	15	550	10	
C40	15% 5K550	5,000	15	750	10	
C41	10% 10K750	10,000	10	550	5	IPDA
C42	10% 10K550	10,000	10	750	5	
C43	10% 5K750	5,000	10	550	5	
C44	10%5K550	5,000	10	750	5	
C45	15% 10K750	10,000	15	550	10	
C46	15%10K550	10,000	15	750	10	
C47	15% 5K750	5,000	15	550	10	
C48	15% 5K550	5,000	15	750	10	

Table 2.3. Formulations of the coatings used in this study (continued).

C49	10% 5K550	5,000	10	550	5	
C50	10%5K750	5,000	10	750	5	
C51	10% 10K750	10,000	10	550	5	
C52	10% 10K550	10,000	10	750	5	
C53	15% 10K750	10,000	15	550	10	1,3-BAC
C54	15%10K550	10,000	15	750	10	
C55	15% 5K750	5,000	15	550	10	
C56	15% 5K550	5,000	15	750	10	
C57	10% 10K750	10,000	10	550	5	
C58	10% 10K550	10,000	10	750	5	
C59	10% 5K750	5,000	10	550	5	
C60	10%5K550	5,000	10	750	5	EPIKURE™ 3233
C61	15% 10K750	10,000	15	550	10	
C62	15%10K550	10,000	15	750	10	
C63	15% 5K750	5,000	15	550	10	
C64	15% 5K550	5,000	15	750	10	

2.2.8. Mechanical properties characterization

The hardness of the coating was evaluated by pencil hardness test according to ASTM D3363 and König Pendulum Hardness according to ASTM D4366. The MEK double rubs examined the coatings' chemical resistance test under ASTM D5402. The impact and conical mandrel tests determined the coatings' flexibility, following ASTM D2794 and ASTM D522, respectively. Tape pull-off adhesion was carried out according to ASTM D3359, where the adhesion of the coating to the steel panel was measured.

2.2.9. Surface characterization

Water and diiodomethane contact angles were measured using a Kruss® DSA 100 (Drop Shape Analyzer) to determine the surface wettability and the surface energy of the coatings. The potential surface interaction with the water droplet over time was studied by measuring the static contact angle over 6 minutes, where a 2- μ L water droplet was monitored on a coating surface.

Also, a tilting method was employed to study the slip angle. The advancing and receding water contact angles for each surface were measured by tilting the stage at 10°/min, where a 25-μL water droplet was monitored on the surface. All contact angle and surface energy results for each coating was auto-calculated using Kruss® Advance software calculated following the Owens-Wendt method.

The surface elemental composition of each coating was examined by A Thermo Scientific™ K-Alpha™ X-ray photoelectron spectroscopy (XPS) instrument, which was equipped with a monochromatic Al K_α (1486.68 eV) X-ray source and an Ar⁺ ion source (up to 4000 eV). A cleaned sample with size 2 mm x 2 mm was sputtered with a large Ar⁺ ion cluster with a power of 4000 eV using the MAGCIS® cluster gun before analysis for each coating. Spectra were collected at an angle normal to the surface (90°) of a 400-μm area at a maintained below 1.5 x 10⁻⁷ Torr chamber pressure ambient temperature. The instrument's software quantified atomic concentrations of C1s, N1s, O1s, and Si2p.

All atomic force microscopy images were generated using a Jupiter XR atomic force microscope (Asylum Research, Oxford Instruments, Santa Barbara, CA, U.S.A.) in Blue Drive air tapping mode using an AC240-TSA-R3 probe. Images of heights and phases were determined. Collecting images on a sample area of 100 μm x 100 μm of the surface of all coatings were scanned (scan rate: 0.7 Hz; point & lines: 512; setpoint: 1.2v; integral gain:60; drive amplitude: 12.63mW; drive frequency: 63.48kHz; tip material: silicon;).

2.2.10. Biological laboratory assays

2.2.10.1. Bacterial (*Cellulophaga lytica*) biofilm adhesion

Bacterial analysis was carried out after 28 days of water immersion preconditioning. Leachate toxicity was assessed by introducing the bacterium into overnight extracts (artificial

seawater with nutrients) of each coating and evaluating growth after 24 hours via a crystal violet colorimetric assay. Growth in coating leachates was reported as an absorbance ratio (600 nm) to growth control. The analysis also included a series of negative growth controls (medium + bacteria + triclosan). *C. lytica* biofilm retention analysis was assessed by a crystal violet colorimetric assay. A 5% suspension of *C. lytica* in ASW + nutrients ($\sim 10^7$ cells.ml⁻¹) were prepared, and 1 mL was added to each well of the coating plate. Plates were incubated statically at 28 °C for 24 hours to facilitate bacterial attachment and colonization. Plates were rinsed three times with DI water and stained with crystal violet. Images were taken after staining, then the crystal violet was extracted in 33% acetic acid (AA), and the resulting eluates were measured for absorbance at 600nm. Biofilm retention was reported as the mean absorbance value of three replicate samples. Error bars represent one standard deviation of the mean. Water jet adhesion was carried out after 24 hours of bacterial biofilm growth. The first column of each plate was not treated and served as the measure of biofilm growth before water jetting. The second and third column of each coating was jetted for 5 seconds at a pressure of 10 psi and 20 psi, respectively. Biofilm adhesion was reported as a function of biomass remaining on the material surface after treatment with each pressure indicated above. Error bars represent one standard deviation of the mean.^{13,19}

2.2.10.2. Growth and release of fouling microalgae, *Navicula incerta*

All microalgae analysis was carried out after 28 days of water immersion. The growth of *Navicula incerta* was assessed by introducing algae into overnight extracts (artificial seawater with nutrients) of treatment and evaluating growth after 48 hours via fluorescence of chlorophyll. Growth in coating leachates was reported as a fluorescence ratio to a growth control (fresh nutrient medium). A negative growth control (medium + bacteria + triclosan) was also included

in the analysis. The initial cell attachment of algae was assessed before water jet adhesion analysis. Algae were diluted to an OD of 0.03 at 660 nm in artificial seawater (ASW) supplemented with nutrients. 1 mL was added to each well of the plate and allowed to incubate statically for 2 hours to facilitate cell attachment. Algal cell attachment was quantified by fluorescence measurement of DMSO extracts of chlorophyll. Cell attachment was reported as fluorescence intensity (relative fluorescence units). Error bars represent one standard deviation of the mean. Water jet adhesion was carried out after 2hr initial cell attachments. The first column of each plate was not treated and served as the measure of cell attachment after 2 hr. The second and third column of each coating was jetted for 10 seconds at a pressure of 10 psi and 20 psi, respectively. Algal adhesion was reported as a function of biomass remaining on the material surface after treatment with each pressure indicated above. Error bars represent one standard deviation of the mean.¹⁹⁻²⁰

2.2.10.3. Adult barnacle 2-week reattachment and adhesion (A. amphitrite)

Coated panels (4" x 8" aluminum) were preconditioned/aged for 28 days via water immersion before the barnacle attachment assessments were conducted. Then six barnacles were immobilized to the surface of the coatings. Coatings were analyzed after 2 weeks of attachment with daily brine shrimp feedings. A hand-held digital force gauge, mounted to an automated stage, was used to measure the peak force of release for each barnacle (lbs.). The number of barnacles that could attach to each surface was recorded, and the attachment efficiency was calculated (percent). Adhesion strength in shear was calculated by dividing the measured force required to remove the barnacle by the basal area and reported in Megapascals (MPa). Each data point is the mean value of the total number of barnacles that are attached to the coating surface. Error bars represent one standard deviation of the mean.²¹⁻²²

2.3. Results and Discussion

Amphiphilic coatings have shown great promise, with good antifouling (AF) and fouling release (FR) performance and robust surfaces. In the previous work of our group, a new novel isocyanate-free coating system was developed that still benefits from urethane linkages. Due to its containing amphiphilic domains, this coating system has good antifouling (AF) and fouling release (FR) performance. In the previous coating system, two different amine curing agents, PACM and Ancamide 702B75 (polyamide resin), were used to cross-link the epoxy group in the coating system, and their effect on performance was explored. This work investigated the effects of additional types of curing agents on the system and variations in PDMS and PEG content.

The glycidyl-carbamate (GC) resin containing epoxy groups was prepared by the isocyanate IPDI trimer and glycidol reaction (Scheme 2.1). The GC prepolymers containing amphiphilic domains were prepared similarly by the reaction between IPDI trimer, glycidol, hydroxyl-terminated PDMS, and hydroxyl-terminated PEG (Scheme 2.2). Molecular weights of PDMS (5,000 $\bar{M}_n$ and 10,000 $\bar{M}_n$) and PEG (550 $\bar{M}_n$ and 750 $\bar{M}_n$) were assessed at two levels in the synthesis of the GC prepolymer, a total of four GC prepolymers were synthesized. The effect of different curing agents, such as polyetheramine, diamine, and alicyclic amine, on performance, was investigated. A total of eight curing agents, shown in Table 2.2, were used in this work, and for each curing agent, formulated with the four different GC prepolymers, the exploring the different amounts of PEG and PDMS (10 wt.%, 15 wt.%). A total of eight formulations were prepared for each curing agent, respectively, and a total of 64 formulations were designed as shown in Table 2.3.

The success of the synthesis of GC resin and GC prepolymers was confirmed by the complete reaction of the isocyanate-functional IPDI trimer with the hydroxyl-terminated

compounds, which was indicated in the FTIR spectrum (Figure 2.1). The absence of the signature strong isocyanate peak at 2250 cm^{-1} and the appearance of the secondary amine (due to the formation of the urethane linkage) and the peaks at 3350 cm^{-1} and 1128 cm^{-1} , respectively, of the GC resin and GC prepolymers, indicated the completion of the reaction. The overlapping stretching for PDMS at 1030 cm^{-1} and PEG at 1105 cm^{-1} in the GC prepolymer spectrum showed the presence of amphiphilic chains in the GC prepolymers.

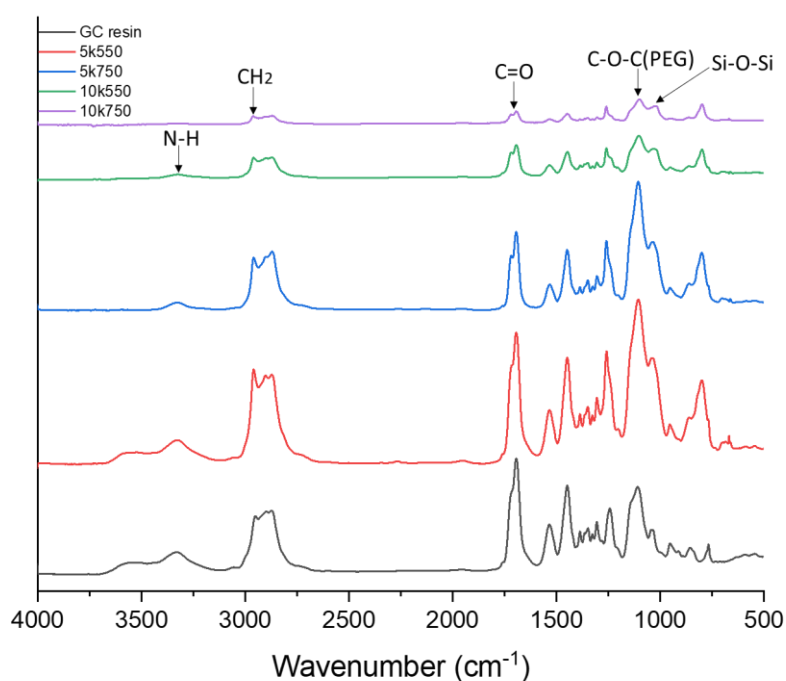


Figure 2.1. FTIR Spectrum for resins: GC resin, GC polymer 5K550, GC prepolymer 5K750, GC prepolymer 10K550, and GC prepolymer 10K750.

The epoxy equivalent weight (EEW) values (Table 2.4) for the GC resin and GC prepolymers by epoxy titrations validated the presence of the epoxy-functional groups. As expected, the titrated EEW of GC resin was close to the theoretical value. The difference between the titrated EEW and the theoretical value of GC prepolymer may be caused by the measurement error due to the high EEW value and relatively small amounts of the specimen. The

EEW of GC resin was much lower than the GC prepolymer, was because of the higher amount of glycidol used. High EEW limits the content of GC prepolymers in the coatings because the use of GC prepolymers will reduce the crosslink density of the coating system.

Table 2.4. Epoxy equivalent weight results.

Materials	Theoretical EEW. g/eq	Titrated EEW, g/eq
GC resin	328.38	321.87
GC prepolymer 10K PDMS-750PEG	3699.79	2677.51
GC prepolymer 10K PDMS-550PEG	2944.05	2139.57
GC prepolymer 5K PDMS-750PEG	3517.70	2711.71
GC prepolymer 5K PDMS-550PEG	2890.68	2245.59

The mechanical properties of the coatings were closely related to the curing agent. For example, Jeffamine D230 and Jeffaimne D400 have the same structure, but the chain of Jeffamine D230 is shorter than Jeffamine D400, so the coating cured with D230 was harder than the coating cured with Jeffamine D400 (shown in Table 2.5). The hardness of the coatings cured with Priamine series curing agents was very low may be due to the low crosslink density and/or possibly incomplete reaction due to their poor compatibility. The same reason also explains why the coatings cured with Priamine series curing agents had poorer MEK double rubs performance than other curing agents especially Priamine_1071. On the contrary, the hardness of the coatings cured with IPDA and 1,3-BAC was much higher (shown in Table 2.5) due to the alicyclic structure. Although EPIKURE™ 3233 also has a long-chain structure, its trifunctional structure increased the crosslink density of the coatings. It improved the hardness of the coatings. The results also showed that the flexibility of the coatings has an opposite trend to the hardness. Also, it is worth noting that most of the crosshatch adhesion tests could not be measured because the coating surface was very oily, and the tape could not adhere to the surface.

Table 2.5. Mechanical properties of coatings.

Formulation	Pencil hardness	Impact test		MEK Double Rubs	Conical Mandrel	Crosshatch Adhesion Test*	Curing agent
		Forward	Reverse				
C1	6H	10	Fail	>400	Fail		
C2	2H	6	Fail	>400	Fail		
C3	3H	5	Fail	>400	Fail		
C4	4H	3	Fail	>400	Fail		Jeffamine D230
C5	2H	7	Fail	>400	Fail		
C6	2H	7	Fail	>400	Fail		
C7	2H	5	Fail	>400	Fail		
C8	2H	6	Fail	>400	Fail		
C9	6H	>43	>43	49	Pass		
C10	F	19	5	>400	Fail		
C11	H	18	3	>400	Fail		
C12	H	29	15	>400	Fail		
C13	HB	>43	34	180	Pass		Jeffamine D400
C14	HB	>43	24	>400	Pass		
C15	H	>43	30	270	Pass		
C16	HB	17	6	>400	Pass		
C17	2H	2	Fail	145	Fail		
C18	H	2	Fail	80	Fail		
C19	2H	2	Fail	29	Fail		
C20	HB	7	Fail	130	Fail		
C21	HB	10	Fail	41	Fail		Priamine_107 1
C22	HB	4	Fail	28	Fail		
C23	2B	4	Fail	40	Fail		
C24	2B	Fail	Fail	63	Fail		
C25	B	3	Fail	240	Fail		
C26	HB	4	Fail	270	Fail		
C27	HB	2	Fail	135	Fail		
C28	B	4	Fail	270	Fail		
C29:	B	4	Fail	52	Pass		Priamine_107 4
C30:	HB	5	Fail	78	Pass		
C31	3B	2	Fail	42	Pass		
C32	3B	4	Fail	42	Fail		

Table 2.5. Mechanical properties of coatings (continued).

C33	HB	2	Fail	210	Fail	
C34	H	F	Fail	183	Fail	
C35	HB	2	Fail	167	Fail	
C36	HB	2	Fail	250	Fail	
C37	HB	4	Fail	85	Fail	Priamine_107 5
C38	HB	4	Fail	109	Fail	
C39	HB	F	Fail	69	Fail	
C40	HB	4	Fail	96	Fail	
C41	3H	1	Fail	>400	Fail	
C42	3H	1	Fail	>400	Fail	
C43	H	1	Fail	>400	Fail	
C44	H	2	Fail	>400	Fail	
C45	2H	2	Fail	>400	Fail	IPDA
C46	2H	1	Fail	>400	Fail	
C47	2H	2	Fail	>400	Fail	
C48	H	2	Fail	>400	Fail	
C49	2H	3	Fail	>400	Fail	
C50	H	2	Fail	>400	Fail	
C51	2H	3	Fail	>400	Fail	
C52	2H	4	Fail	>400	Fail	
C53	3H	5	Fail	>400	Fail	1,3-BAC
C54	3H	3	Fail	>400	Fail	
C55	3H	4	Fail	>400	Fail	
C56	3H	2	Fail	>400	Fail	
C57	2H	5	Fail	>400	Fail	
C58	H	5	Fail	>400	Fail	
C59	2H	4	Fail	>400	Fail	
C60	2H	5	Fail	>400	Fail	EPIKURE™ 3233
C61	2H	14	1	>400	PASS	
C62	2H	13	1	>400	Fail	
C63	2H	9	1	>400	PASS	
C64	2H	7	Fail	>400	Fail	

*Crosshatch adhesion test did not complete successfully due to the oily surface.

Static contact angle measurements of water and diiodomethane were done at zero time and after 6 minutes to monitor differences due to the changes in the coating surfaces over time

(Fig. 2.2.-Fig. 2.5.). Surface energy for each coating was calculated using the Owens-Wendt method. Most of the water contact angle values of the coatings were greater than 90 degrees, suggesting the hydrophobic character of the surface. The water contact angle was mainly affected by the molecular weight of the PDMS used. The water contact angle of the coating using 10K molecular weight PDMS was higher than that using 5K PDMS, showing higher hydrophobicity. Most of the contact angles of the coatings also increased when PEG with a higher molecular weight was used. Under the same conditions, the contact angle difference between the coatings containing 10 wt.% PEG/PDMS and the coatings containing 15 wt. % PEG/PDMS content was insignificant.

The coatings cured with alicyclic amine had a higher water contact angle than those cured by other curing agents when containing the same PDMS and PEG content, indicating higher hydrophobicity, which may be attributed to their lower hydrophilicity compared with polyetheramine. For example, for the coatings cured with IPDA, the contact angle of the coating using 10% GC prepolymer synthesized with 5K PDMS and 550 PEG was lower than that of the coating using 10% GC prepolymer synthesized with 5K PDMS and 550 PEG, and the contact angle of the coatings using GC prepolymer synthesized with 10K PMDS was overall greater than that synthesized with 5K PDMS. Correspondingly, the surface energy of the coating had an opposite trend. Most of the surface energy decreased with the increase of the molecular weight of PEG in a GC prepolymer with the same PDMS. Both happened when the PDMS/PEG content was ten wt.% and 15 wt.%. Consider the coatings cured with IPDA as an example. The coatings with 10% GC synthesized with 5K550 had the highest surface energy. In contrast, the coatings cured with the hydrophobic curing agent Priamine series present a variable surface, mainly

because the Priamine series curing agent was incompatible with the system, and the cured film was not uniform.

The water contact angle of the coatings cured with Jeffamine D230 was generally higher than those cured with Jeffamine D400, which also indicated that the denser the film, the higher the water contact angle of the film. All the coatings exhibited a dynamic surface, with a decreasing trend in the contact angle (both water and diiodomethane) over time, which was attributed to the rearrangement of the surface caused by the presence of PEG and PDMS domains, leading the films to swell in both polar and non-polar environments. Formulations with 15 wt.% PEG and PDMS showed more significant dynamic changes than formulations with 10 wt.% PEG and PDMS.

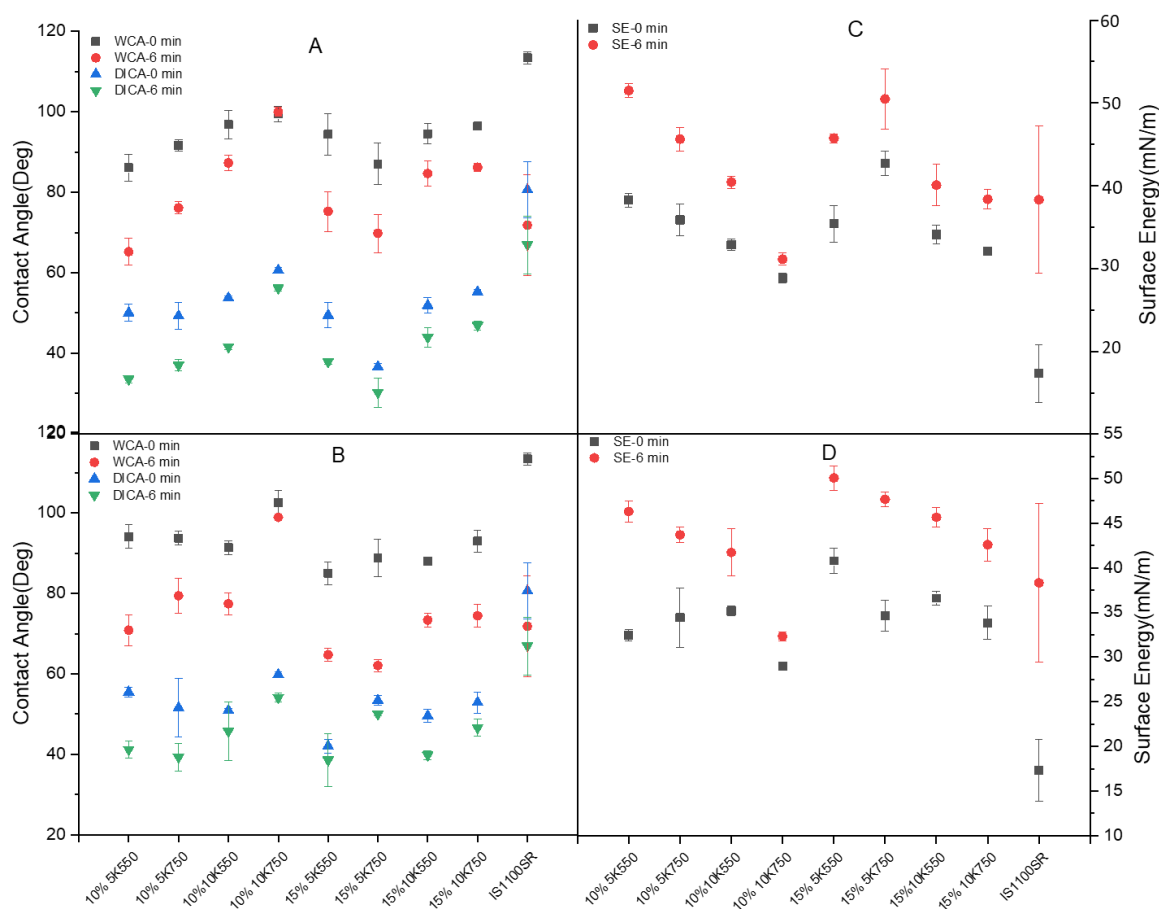


Figure 2.2. Static contact angle data for coatings: A-D are water contact angles (WCA) and diindolylmethane contact angles (DICA) at 0 minutes and 6 minutes, and their surface energy of coatings cured with Jeffamine D230 and with Jeffamine D400 separately, the x-axis is labeled to specify the formulations and its components, including PEG MW, PDMS MW, wt.% of PEG and PDMS. The error bars are standard deviations of three measurements.

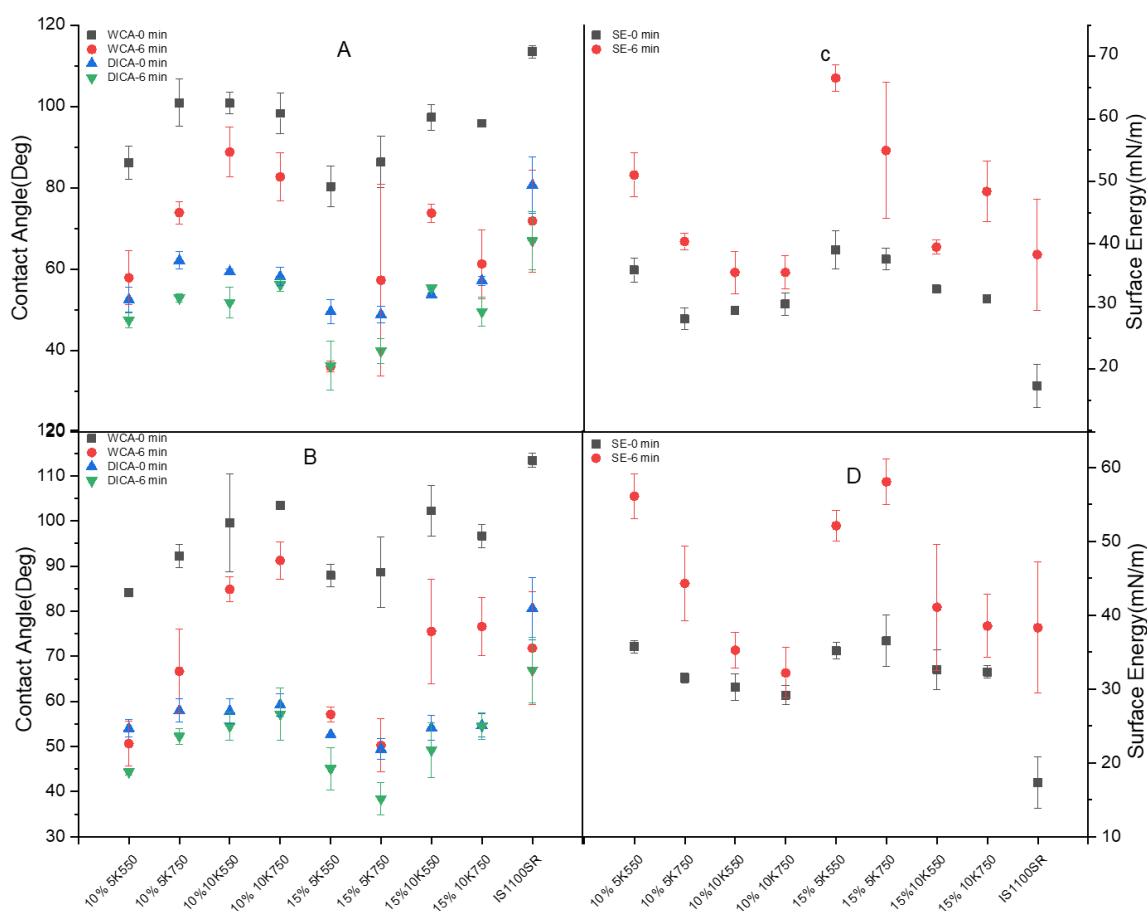


Figure 2.3. Static contact angle data for coatings: A-D are water contact angles (WCA) and diindolylmethane contact angles (DICA) at 0 minutes and 6 minutes, and their surface energy of coatings cured with Priamine_1071 and with Priamine_1074 separately, the x-axis is labeled to specify the formulations and its components, including PEG MW, PDMS MW, wt.% of PEG and PDMS. The error bars are standard deviations of three measurements.

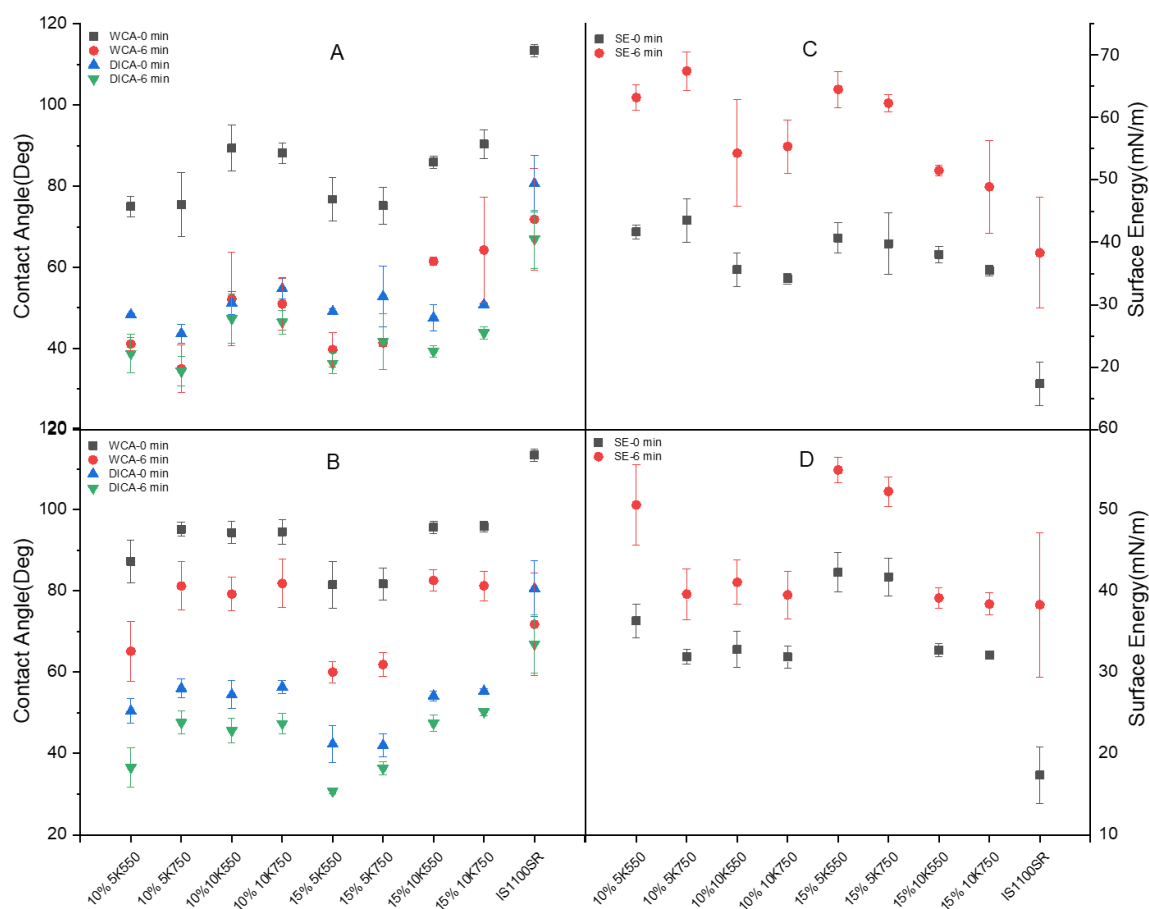


Figure 2.4. Static contact angle data for coatings: A-D are water contact angles (WCA) and diindolylmethane contact angles (DICA) at 0 minutes and 6 minutes, and their surface energy of coatings cured with Priamine_1075 and with EPIKURE 3233 separately, the x-axis is labeled to specify the formulations and its components, including PEG MW, PDMS MW, wt.% of PEG and PDMS. The error bars are standard deviations of three measurements.

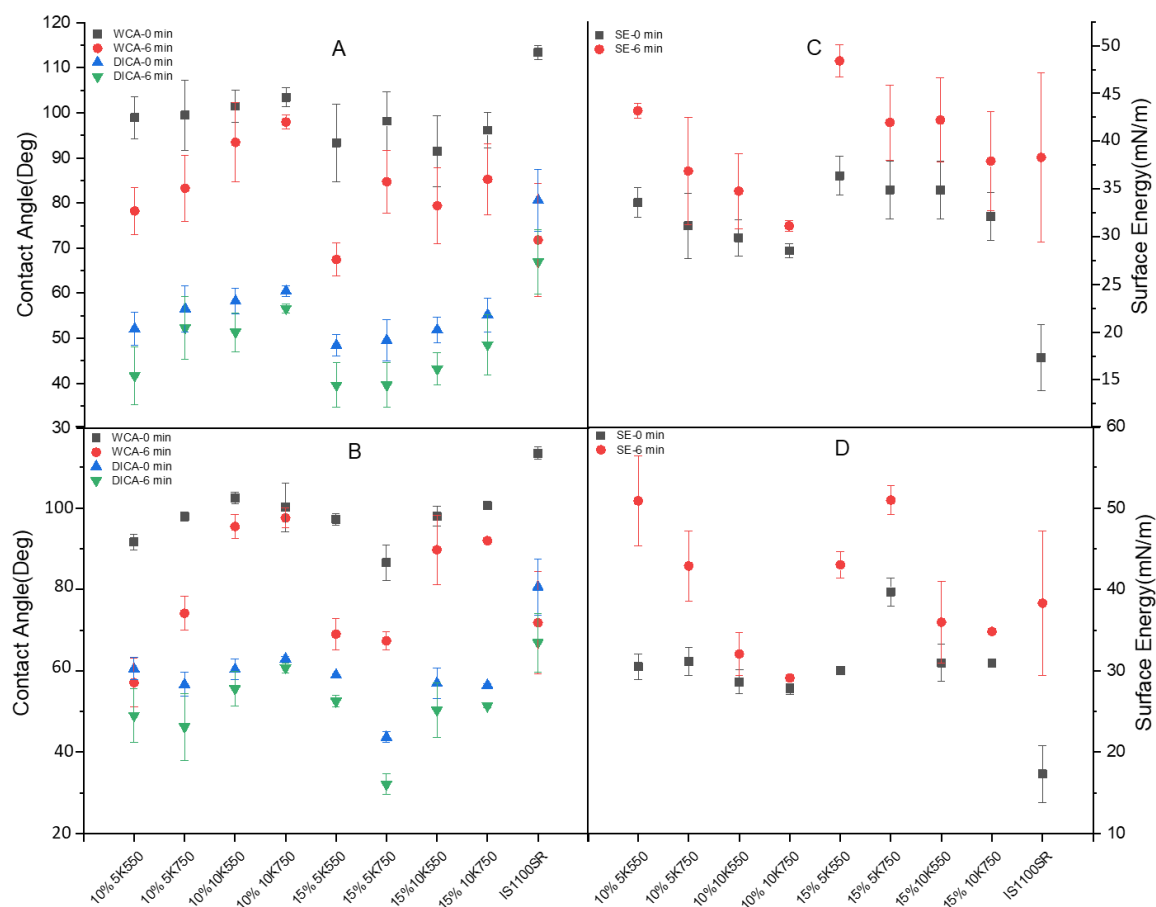


Figure 2.5. Static contact angle data for coatings: A-D are water contact angles (WCA) and diindolylmethane contact angles (DICA) at 0 minutes and 6 minutes, and their surface energy of coatings cured with IPDA and with 1,3-BAC separately, the x-axis is labeled to specify the formulations and its components, including PEG MW, PDMS MW, wt.% of PEG and PDMS. The error bars are standard deviations of three measurements.

Dynamic contact angle measurements were further used to study the surfaces of the coatings to determine advancing/receding contact angles and contact angle hysteresis (Fig. 2.6.). The contact angle hysteresis had a consistent trend with their surface energy. Overall, most formulations had low hysteresis values, implying hydrophobicity. Still, several formulations had higher hysteresis values, for example, coatings made with the Jeffamine D400, 1,3-BAC, and Priamine_1071 curing agents containing 10 wt.% and 15 wt.% prepolymers 5K550, indicating the surface is not as smooth as others which may be attributed to the defects or differences in

rearrangements chemistry of the surface. Also worth noting is that under the same GC prepolymer content, the coating using GC prepolymer synthesized with low molecular weight PEG (550 $\bar{M}_n$) and PDMS (5K $\bar{M}_n$) had the highest contact angle hysteresis, which means that when the lower molecular weight PEG (550 $\bar{M}_n$) was used, the PEG domain seems to reach the surface more easily, making the surface more hydrophilic, which will be further confirmed in later tests.

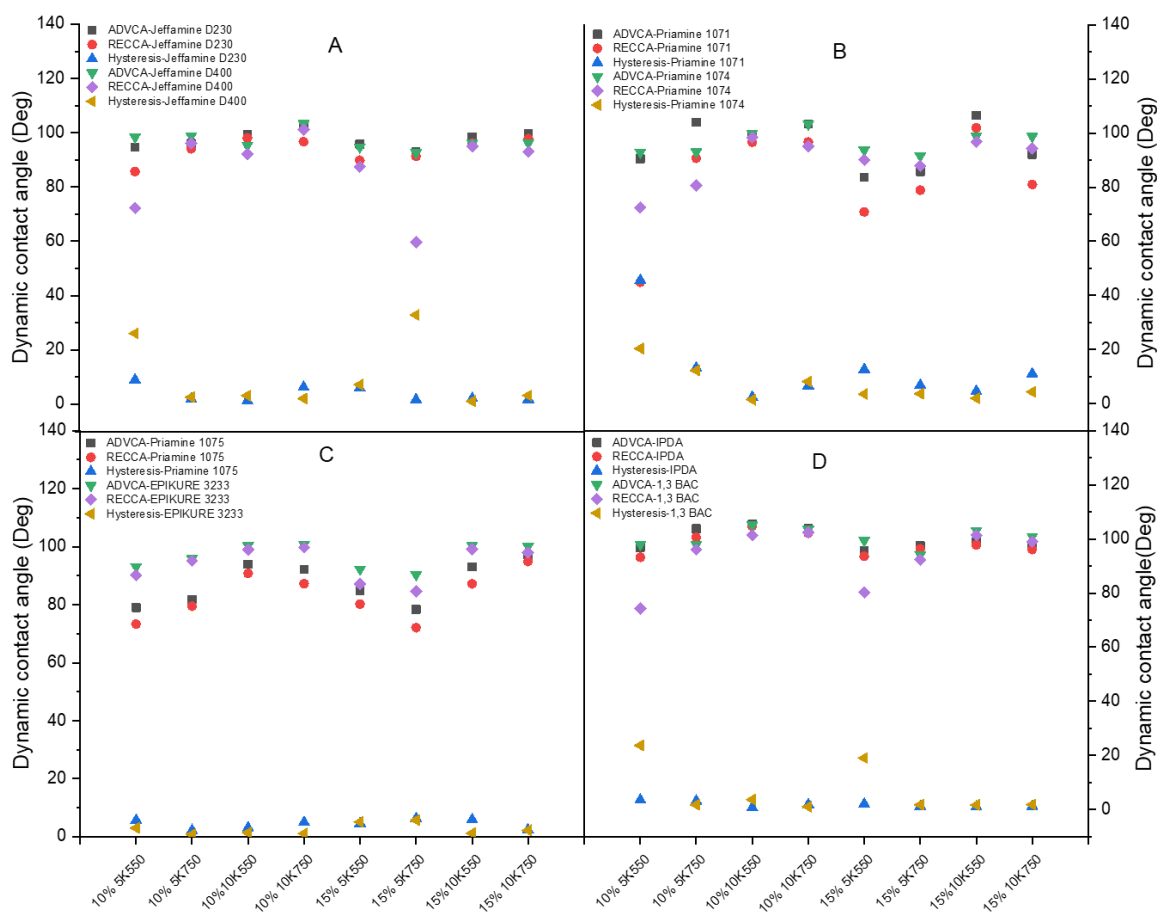


Figure 2.6. Dynamic contact angle data for AmpSiGC coatings: A-D advancing (ADVCA) and receding (RECCA) water contact angles (WCA) and contact angle hysteresis of coatings cured with Jeffamine D230 to EPIKURE 3233 separately, the x-axis is labeled to specify the formulations and its components, including PEG MW, PDMS MW, wt.% of PEG and PDMS.

XPS was performed on the 64 coatings used to observe the atomic content on and just below the surface, indicating amphiphilicity (Fig. 2.7.). Generally speaking, all of the coatings

containing higher molecular weight PDMS (10K $\bar{M}_n$) had a higher Si atomic content on the surface of coatings indicated by the XPS analysis, but the trend was not significant, which may be because the effect of PDMS molecular weight on the Si content on the coating surface leads to a plateau. After reaching a specific molecular weight, the trend of this effect decreases with the increase in molecular weight, which will be clearly shown in the following research. On the other hand, for the same GC prepolymer used, the higher amount in the coating formulation, the higher the concentration of silicon atoms on the coating surface. It was worth noting that when the content of GC prepolymer was fixed, the surface silicon atoms percentage of the coating using GC prepolymer, which was synthesized with 5K PDMS and 750 PEG, was higher than using GC prepolymer which was synthesized with 5K PDMS and 550 PEG, which PEG had the lowest surface silicon atoms content at a fixed GC prepolymer content. But the opposite result can be seen that the surface silicon atoms percentage of the coating using GC prepolymer, which was synthesized with 10K PDMS and 750 PEG, was lower than using GC prepolymer, which was synthesized with 10K PDMS and 550 PEG. This is because PDMS and PEG have opposite effects on the surface self-assembly of coatings, and their molecular weights determine this effect. They will change who occupies the dominant position. Most formulations using 10K PDMS have a Si atoms content between 15-25 wt.% on the surface, compared to less than 15% Si atoms content on the surface of formulations using 5K PDMS. However, there is a noteworthy phenomenon that, for the same part A, the Si content on the surface of the coating is higher when using a potent polar curing agent (such as Jeffamine D400). This is because the system's polarity was enhanced when using an opposing curing agent, which increases the repulsion of PDMS. The force makes it easier for PDMS to reach the coating surface.

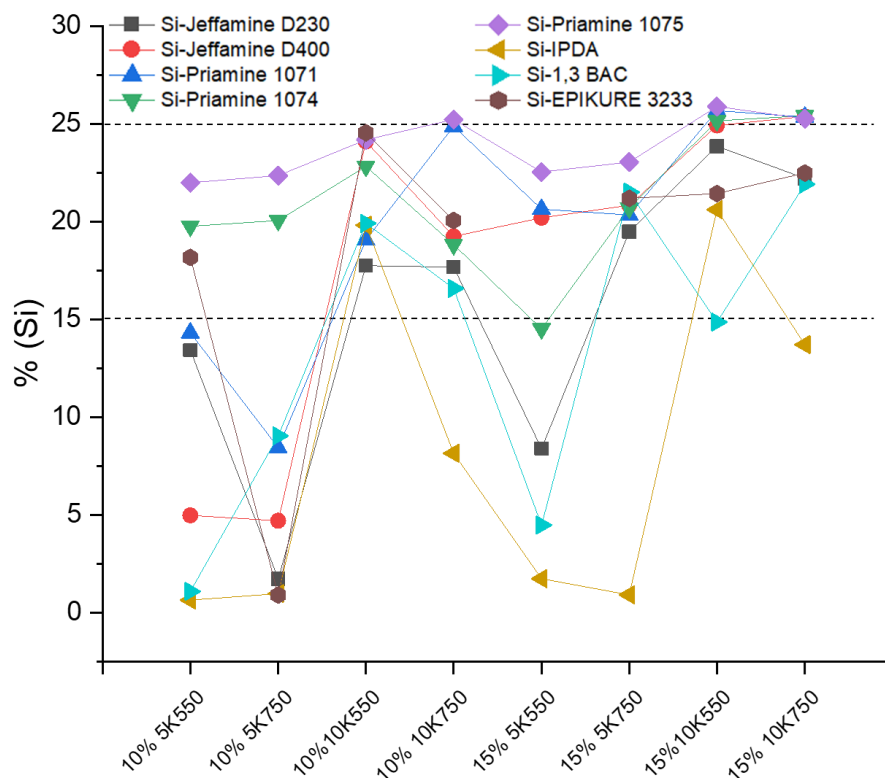


Figure 2.7. XPS data: Si atomic content on and just below the surface, indicating the effect of molecular weight of PMDS and PEG on surface self-assembly of coatings. The effect of curing agents on the silicon atom content on the surface suggested that the polarity of curing agents played a significant role in self-stratification and surface composition.

To observe the surface microstructure, atomic force microscopy (AFM) was performed on the coatings. The height images and corresponding phase images of the coating cured with Priamine_1075 were obtained and were shown below (Fig. 2.8.) as examples. For each formulation, the top image is the height image, and the bottom image is the phase image. The formation of rough topography and the height variation is caused by the microphase segregation of the hydrophobic segments PDMS and hydrophilic components PEG on the surface. The smaller phase angles indicate a harder domain (PEG). Higher phase angles indicate the presence of a softer domain (PDMS) in this case. When the content of PDMS and PEG is 10 wt.%, the

size of the hydrophilic domains (PEG) on the surface of the coating using the GC prepolymer synthesized with small molecular weight PDMS (5K $\bar{M}_n$) was smaller. They seemed more inclined to move away from each other, while when the hydrophilic domain's size on the coating surface using GC prepolymer synthesized by higher molecular weight PDMS (10K, $\bar{M}_n$) was larger. This trend was more apparent when the molecular weight of PEG increased, which was close to aggregation. When the content of PDMS and PEG is 15 wt.%, no matter whether the used GC prepolymer synthesized with 5K $\bar{M}_n$ PDMS or 10K $\bar{M}_n$ PDMS, the hydrophilic domain (PEG) seems to be more difficult to aggregate and can only keep a small size and are well dispersed on the surface. The hydrophilic domains (PEG) on the coating surface using GC prepolymer synthesized with low molecular weight PDMS (5K $\bar{M}_n$) seem more uniformly dispersed.

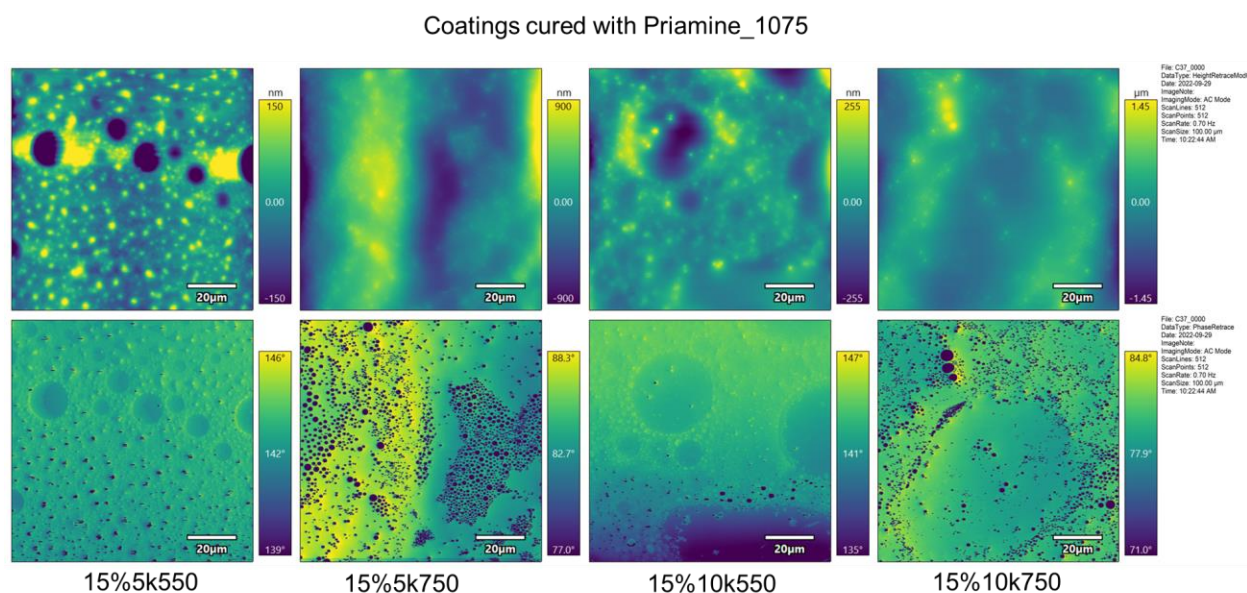
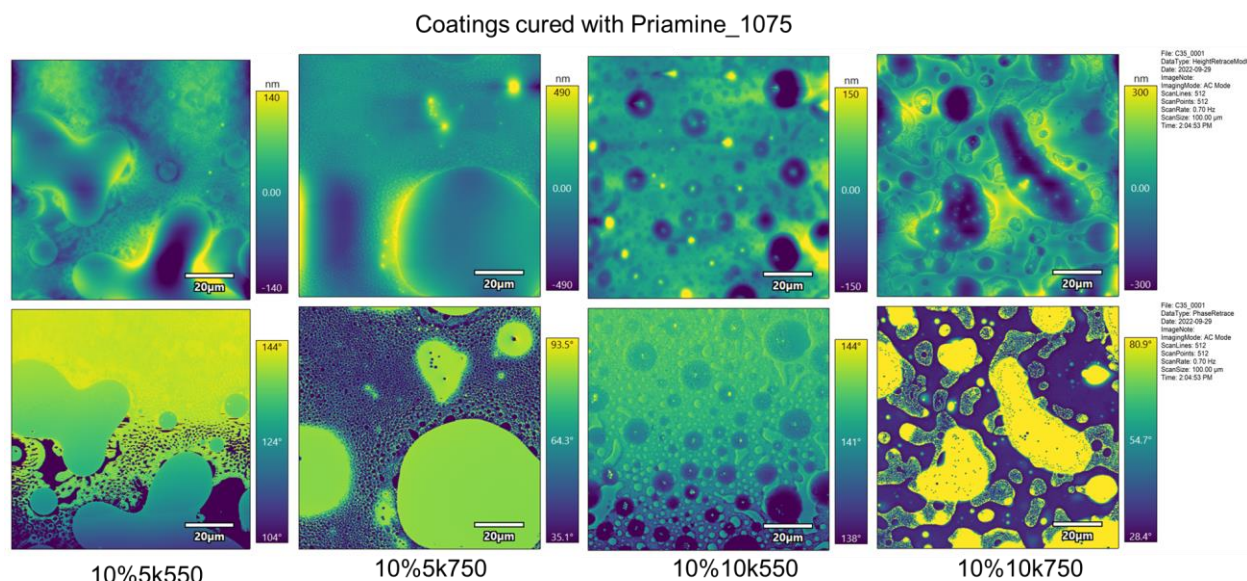


Figure 2.8. AFM Height and Phase images of coatings cured with Priamine_1075, each image represents an area of 100 μm x 100 μm . Each label reflects the coating number and composition of an image.

By immersing all coatings in water for 28 days, a leachate toxicity assessment was completed before the biological assays to assess FR performance were conducted to ensure all coatings were without any toxic ingredients. Diatom *Navicula incerta* and bacterium *Cellulophaga lytica* were used to evaluate the fouling release performance of all coatings in the laboratory. The cell attachment for *N. incerta* and *C. lytica* was stimulated by static incubation

for two hours under ambient conditions in the dark, and coating surfaces were then subjected to water-jet treatments at 10 and 20 psi water pressure for ten seconds. The relative fluorescence (RFU) measured from the extracts was directly proportional to the biomass remaining on the coating surfaces after water-jetting, which was used to evaluate the *N. incerta* and *C. lytica* fouling release performance.

The coatings cured by hydrophobic curing agents (such as Priamine_1075, Figure 2.10. C) had a lower initial extent of diatom biofouling, indicating they had a better anti-fouling (*N. incerta*) performance than that cured by polar curing agents (such as Jeffamine D400, Figure 2.9. B), and even showed a similar anti-fouling performance to that of the control IS1100SR. The release of diatom film was evaluated at 10 psi and 20 psi water pressure levels. The results showed that the coatings cured with hydrophobic curing agents had a better fouling release performance than those cured with polar curing agents. This is because the slime-forming diatom *N. incerta* is another major biofouling organism that was generally thought to tend to attach to hydrophobic surfaces. The coatings cured with polar curing agents had more silicon atoms on the surface, as we discussed above. A similar trend was observed for all coatings cured with different curing agents at both pressures. It was that with a fixed content of GC prepolymer used, the biofilm remaining increased by order of using GC prepolymers synthesized with 5K PDMS-550 PEG, 5K PDMS-750PEG, 10K PDMS-550 PEG, and 10K PDMS- 750 PEG, and the coating using 15 wt.% GC prepolymer synthesized with 5K PDMS-550 PEG had the best *N incerta* release performance both at 10 psi and 20 psi water pressure levels, demonstrated a better fouling release performance than several of the controls, including amphiphilic T-2 and commercial systems, IS 700 systems. The coatings cured with Primaine_1071, Priamine_1074, and Priamine_1075 labeled 10% 10K750, 15%5K750 and 15% 10K750 had a blistered surface

or/and removable layer after leachate toxicity assessment, so they didn't move forward to the cell attachment assessment. The reason that these coatings had a bad surface after immersing in water for 28 days is that these coatings had low crosslink density.

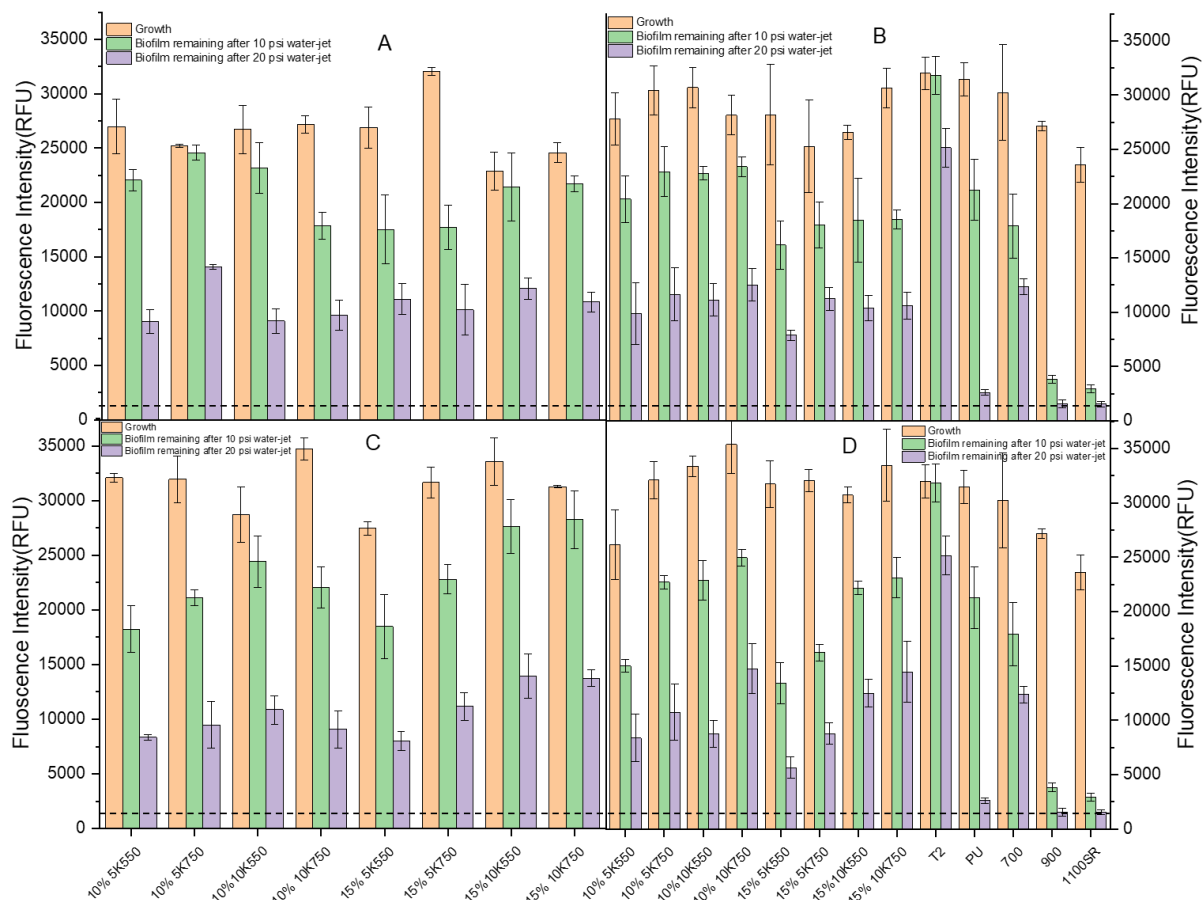


Figure 2.9. *N. incerta* data for biofilm growth (yellow bar), release at 10 psi (green bar), and release at 20 psi (purple bar). The X-axis is labeled to specify the formulations and their components, such as wt.% of PEG and PDMS and controls. A-D are coatings cured with Jeffamine D230, Jeffamine D400, IPDA, and 1,3-BAC, respectively. The error bars are standard deviations of six measurements.

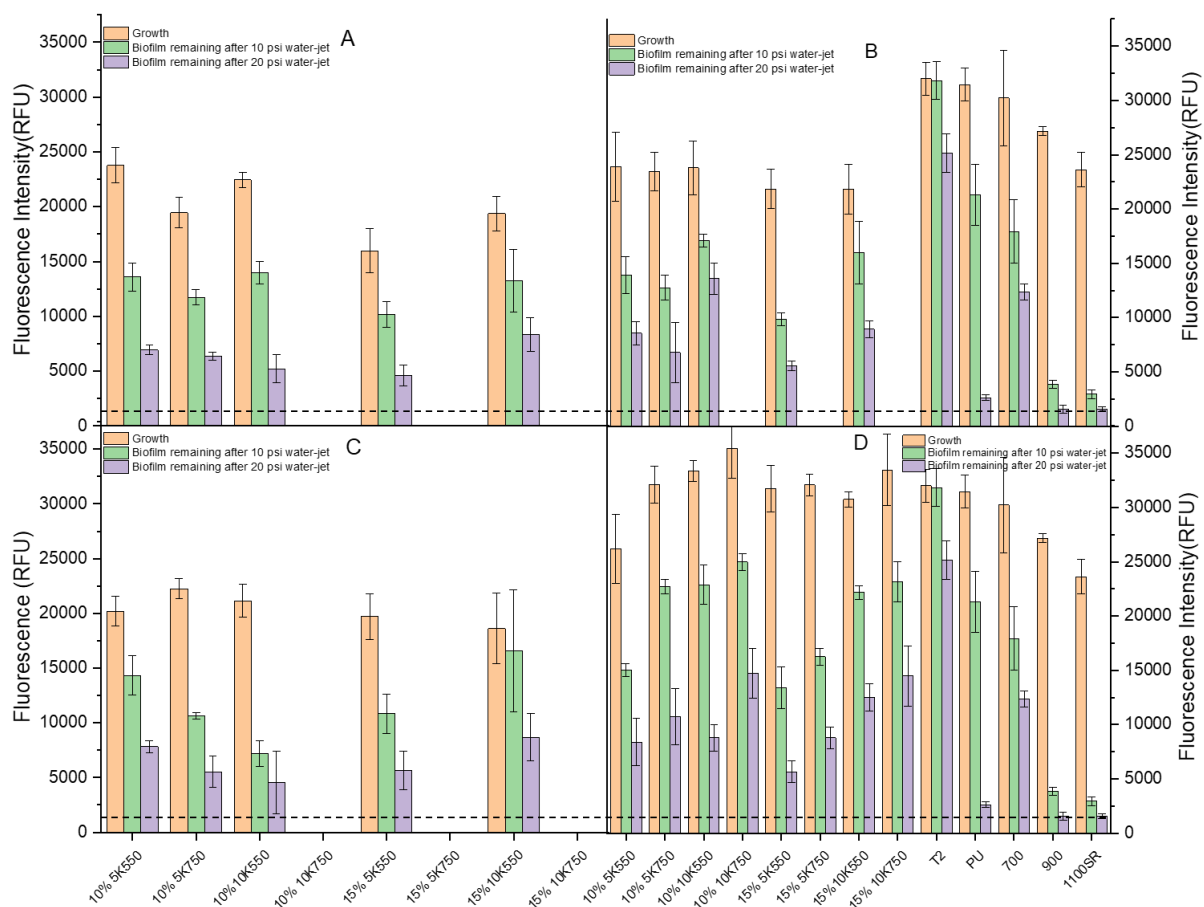


Figure 2.10. *N. incerta* data for biofilm growth (yellow bar), release at 10 psi (green bar), and release at 20 psi (purple bar). The X-axis is labeled to specify the formulations and their components, such as wt.% of PEG and PDMS and controls. A-D are coatings cured with Priamine_1071, Priamine_1074, Priamine_1075, and EPIKURE 3223, respectively. The error bars are standard deviations of six measurements.

Fouling-release properties towards bacteria were evaluated using laboratory biological assays. A suspension consisting of the marine bacterium *C. lytica* at 107 cells/mL concentration in artificial seawater were incubated statically at 28°C for 24 hours, then subjected to water-jet treatments at 10 and 20 psi, respectively. FR performance can be obtained by comparing the absorbance values, which were directly proportional to the amount of bacterial biofilm on coating surfaces before and after water-jetting treatments.

C. lytica is a known biofouling microorganism with an affinity for a wide range of hydrophilic and hydrophobic surfaces. Compared to the controls, the fouling release performance at both 10 and 20 psi water pressure level of most of the coatings in this work were much better than the controls T2, PU, Intersleek 700, and Intersleek 900, and similar or better than Intersleek 1100SR, which because that most of the coatings in this work had very good amphiphilicity. Coatings with 15 wt.% GC prepolymer synthesized with 5K PDMS and 550PEG demonstrated the best fouling release performance at both 10 and 20 psi water pressure. In internal comparison, the coatings containing 10 wt.% GC prepolymer mostly performed better than the coatings containing 15 wt.% except the coatings cured with 1,3BAC, which with 10 wt.% amphiphilic chains did not perform as well as the coatings with 15 wt.%. The coatings cured with Primaine_1071, Priamine_1074, and Priamine_1075 labeled 10% 10K750, 15%5K750, and 15% 10K750 had a poor surface quality after leachate toxicity assessment, so they didn't move forward to the cell attachment assessment. These coatings had poor surface quality after immersing in water for 28 days since these coatings had low crosslink density.

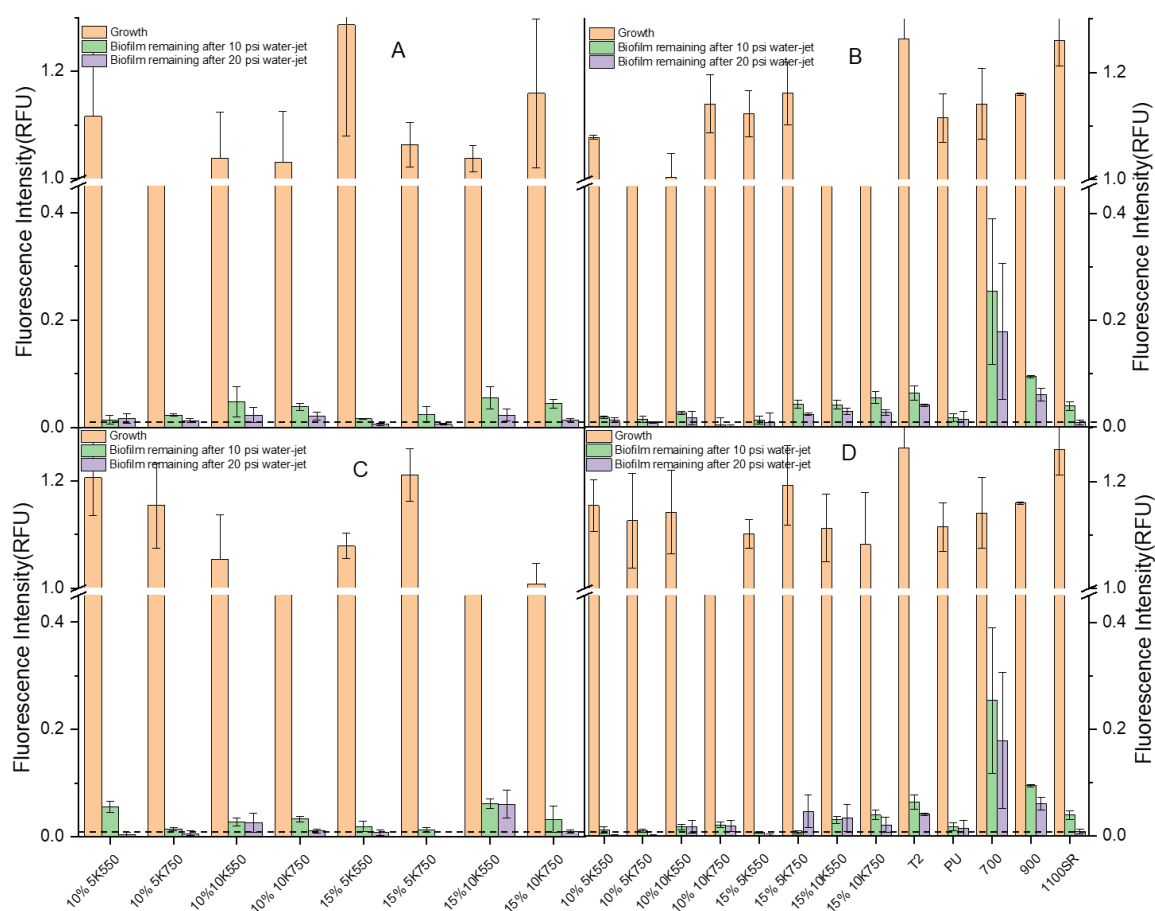


Figure 2.11. *C. lytica* data for biofilm growth (yellow bar), release at 10 psi (green bar), and release at 20 psi (purple bar). The X-axis is labeled to specify the formulations and their components, such as wt.% of PEG and PDMS and controls. A-D are coatings cured with Priamine_1071, Priamine_1074, Priamine_1075, and EPIKURE 3223, respectively. The error bars are standard deviations of six measurements.

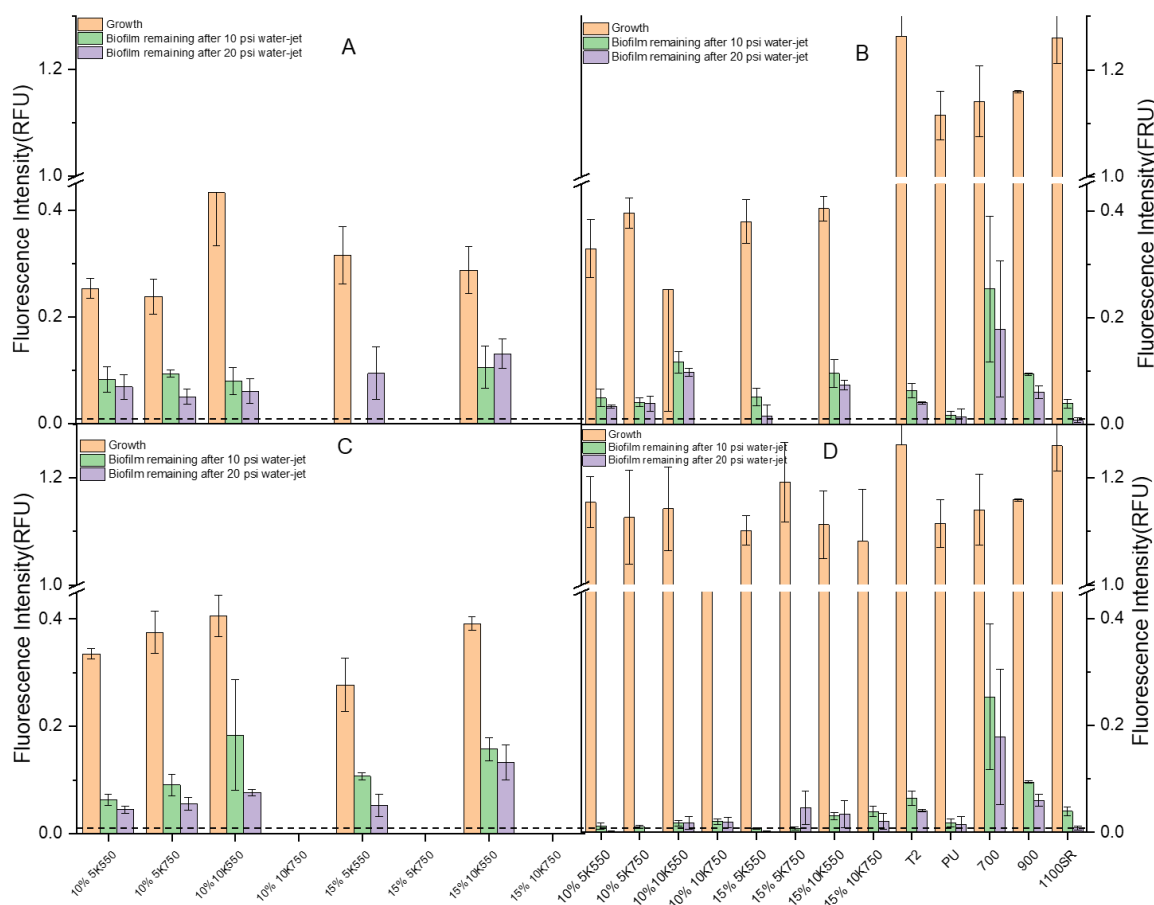


Figure 2.12. *C. lytica* data for biofilm growth (yellow bar), release at 10 psi (green bar), and release at 20 psi (purple bar). The X-axis is labeled to specify the formulations and their components, such as wt.% of PEG and PDMS and controls. A-D are coatings cured with Jeffamine D230, Jeffamine D400, IPD, and 1,3-BAC, respectively. The error bars are standard deviations of six measurements.

Four of the 64 coatings were selected for barnacle adhesion assessment. These four coatings had balanced good biofouling removal performance and covered all coating design features. They were coatings with 15 wt.% concentration of PDMS/PEG of GC prepolymer synthesized by 5K $\bar{M}_n$ PDMS and 550 $\bar{M}_n$ PEG, cured with Jeffamine D400, IPDA, 1,3-BAC, and Epikure 3223, respectively. Figure 2.13. shows the result of the number of barnacles reattached to the coating surface and barnacle adhesion. The number of barnacles reattached to the coatings cured with Jeffamine D400, IPDA, 1,3-BAC were higher than the control IS

1100SR due to the low silicon content on the coating surface (Fig. 2.13.). In contrast, the coating cured with Epikure 3223 had fewer reattached barnacles. However, the barnacle adhesion result didn't show the same trend. The coatings cured with IPDA and 1,3-BAC had low silicon on the surface, but the coating cured with IPDA was more easily released than the coating cured with 1,3-BAC. The same was shown of the coatings cure with Jeffamine D400 and Epikure 3223. Both of them had high silicon content on the coating surface but displayed different release performances, indicating that besides surface wettability, other factors affect the coating's release performance against barnacles. Nevertheless, the results showed that the experimental coatings' barnacle adhesion was within an order of magnitude compared to the commercial coating.

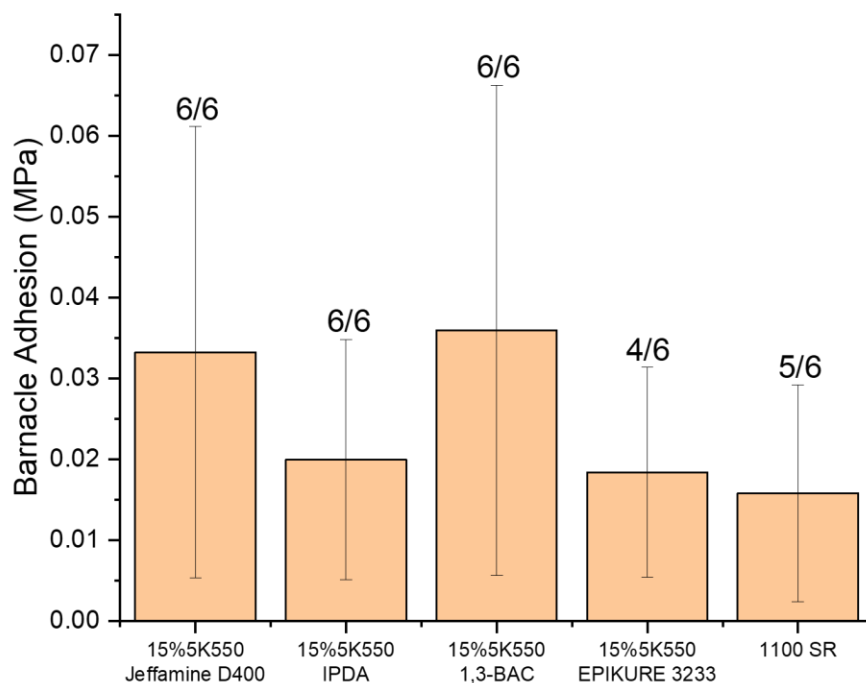


Figure 2.13. Reattached barnacle (*A. amphitrite*) adhesion strength data. Six barnacles were used for each reattachment. The number of attached barnacles out of six is labeled as a ratio for each system. Each bar shows the average adhesion strength based on the number of successfully reattached barnacles. The X-axis is labeled to specify the formulations and their components I, PEG MW, PDMS MW, wt.% of PDMS and PDMS, and curing agent type. The error bars are standard deviations.

2.4. Conclusions

The glycidyl-carbamate resin and glycidyl-carbamate with amphiphilic domains were successfully synthesized by the reaction of IPDI trimer resin with glycidol PDMS, and PEG, converting the NCO functional group to epoxy group, which can reduce health hazards when used by workers. The FTIR spectra confirmed the completed reactions. A total of 64 coatings were prepared in this work to study the factors that affect the anti-fouling and fouling release performance. The factors include the molecular weight of PDMS and its amount in the system, the molecular weight of PDMS and its amount in the system, and the curing agents' type. The

surface study showed that the curing agents and molecular weight of PDMS and PEG played a significant role in the self-assembly behavior of the coatings. Contact angle measurements indicated that amphiphilic coatings have a dynamic surface. XPS results demonstrated the effects of the molecular weight of PDMS, PEG, and their amount in the system on the coatings' self-stratification. The curing agent's polarity also greatly influenced the content of silicon atoms on the coating surface. Under the same conditions, a highly polar curing agent made it easier for PDMS to reach the surface. AFM confirmed the presence of amphiphilic domains on the coating surface. The shape of domains depends much on the molecular of PDMS and PEG, their amount, and the type of coatings. The biological measurements showed that the amphiphilic coatings had promising anti-fouling and fouling release performance. Especially for biofouling microorganisms with an affinity for a wide range of surfaces, both hydraulic and hydrophobic, such as *C. lytica*, the coating in this work had excellent fouling release performance, which is attributed to the high 'confused' amphiphilic surface.

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CHAPTER 3. STUDY OF CRITICAL HYDROPHILIC-LIPOPHILIC RATIO (CHLR) OF THE AMPHIPHILIC GC COATING SYSTEM

3.1. Introduction

In Chapter 2, Alireza Rahimi's study was extended to use the same method to synthesize GC resin and GC prepolymer and use a wider variety of curing agents to investigate the effect of curing agents.¹ The selected curing agents, including the main commercial amine curing agents such as polyether amine dimer diamine, and cycloaliphatic amine, a total of 8 curing agents were investigated. Compared to the controls, the bacteria fouling release performance at both 10 and 20 psi water pressure levels of most of the coatings in this work were much better than the controls T2, PU, Intersleek 700, and Intersleek 900, and similar or better than Intersleek 1100SR, which is due to most of the coatings having very good or even excess amphiphilicity. Meanwhile, the diatom fouling release performance at both 10 and 20 psi water pressure levels of most of the coatings in Chapter 2 work poorer worse than most controls. In this chapter's work, the PDMS and PEG were reacted separately with glycidol and IPDI trimer to synthesize PMDS/PEG GC resins with diepoxy groups so that any combination of PDMS and PEG could be realized. The bacteria and diatom fouling release performance were balanced by adjusting the molar ratios of PDMS to PEG (from 100:1-0.5:1). Results showed that the coatings with the hydrophilic-lipophilic ratio (CHLR) at 1:1 to 2:1 had better anti-fouling and fouling release performance.

3.2. Experimental

3.2.1. Materials

Covestro LLC provided isophorone diisocyanate (IPDI) polyisocyanate trimer Desmodur Z4470 BA. Monocarbinol-terminated polydimethylsiloxanes (CT-PDMS) with molecular

weights of 1000 $\bar{M}_n$ (MCR-C12), 5000 $\bar{M}_n$ (MCR-C18) and 10,000 $\bar{M}_n$ (MCR-C22) were purchased from Gelest, Inc. Methyl ethyl ketone (MEK), methyl amyl ketone (MAK), hydrobromic acid (5.7 M), standardized hydrochloric solution (0.1 N) was purchased from Alfa Aesar Glycidol, poly(ethylene glycol) methyl ether (550 $\bar{M}_n$ and 750 $\bar{M}_n$), and dibutyltin diacetate (DBTDAc) were purchased from Sigma Aldrich. Toluene and glacial acetic acid were purchased from VWR. Curing agent Priamine_1075 was supplied by Croda Inc. Dow Corning provided silicone elastomer Silastic® T2 (T2), and AkzoNobel International Paint provided Intersleek® 700 (IS-700), Intersleek® 900 (IS-900), and Intersleek® 1100SR (IS-1100) as commercial fouling release coatings controls. Intergard 264 was also provided by International Paint and was used to prime aluminum panels (4" x 8" in., 0.6 mm thick, type A, alloy 3003 H14) purchased from Q-lab. Q-lab also provided steel panels (3" x 6" in., 0.2 mm thick, smooth finish) for mechanical tests.

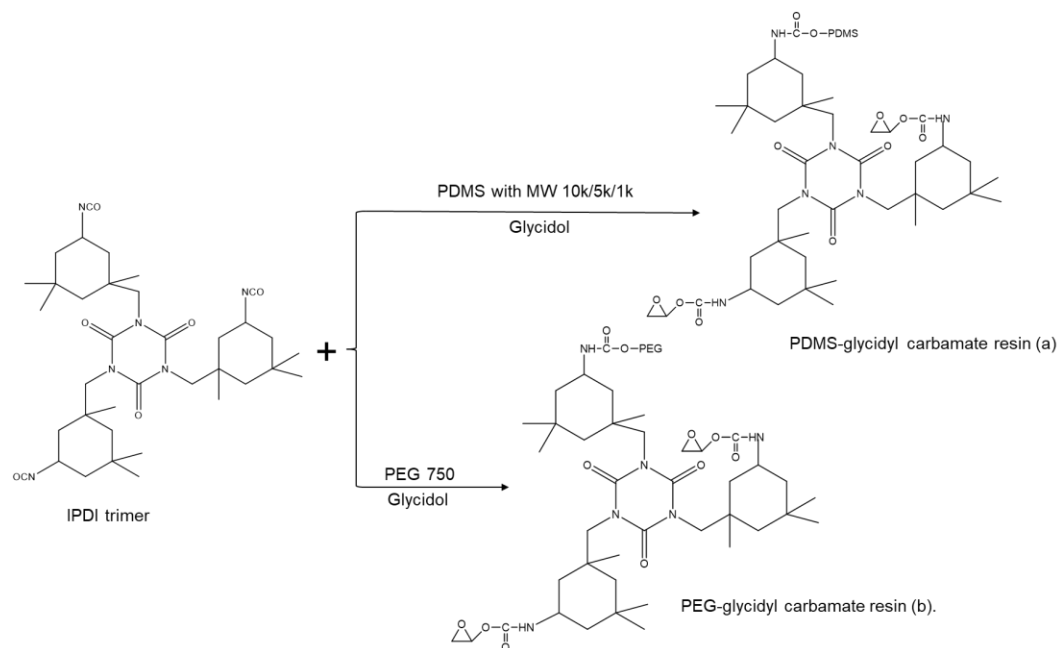
3.2.2. Synthesis of glycidyl carbamate (GC) resin, PDMS-GC and PEG-GC

Glycidyl carbamate (GC) resin was synthesized as described previously by reacting the IPDI trimer resin with glycidol in a 1:1 ratio of -NCO: -OH, as Scheme 2.1 shows. PDMS-glycidyl carbamate resin and PEG-glycidyl carbamate resin were synthesized separately by the reaction between IPDI trimer (Desmodur Z4470), glycidol, and PDMS/or PEG, as shown in Scheme 3.1. In this work, the molecular weight of PDMS was a key factor to be investigated. Three different molecular weights of PDMS (1,000 $\bar{M}_n$, 5,000 $\bar{M}_n$, 10,000 $\bar{M}_n$) and one PEG molecular weight (750 $\bar{M}_n$) were used to synthesize functional glycidyl carbamate resins, respectively. A total of five resins were prepared, as Table 3.1 shows. The ratio of -NCO and -OH was 1:1, and glycidol provided 66.67% -OH groups, and the remaining 33.33% -OPEG or PDMS provided OH groups. For example, to make the 1K PDMS GC resin, 85 g carbinol-PDMS 1K $\bar{M}_n$ (OH eq wt. = 1,000)

was first diluted in toluene (40.83g) in a 250-mL four-neck flask and mixed robustly with vortexing for 2 minutes. Glycidol (13.1 g) was added and mixed for another 2 minutes. Following this, 96.626 g of IPDI trimer resin (NCO eq wt. = 269.53), and 9.36g of DBTDAc catalyst solution (by wt. 1% in MAK) were added to the flask. The reaction was carried out at 80 °C for 2-3 hours. The flask was equipped with a stirrer, nitrogen inlet, reflux condenser, and temperature controller. In theory, the synthesized prepolymer contained 51.08 wt.% PDMS.

Table 3.1. Synthesis of GC resin and amphiphilic functional GC resin.

ID	PDMS MW ($\bar{M}_n$)	PEG MW ($\bar{M}_n$)	Wt.% of PDMS (or PEG) in solid
GC resin	-	-	-
10K PDMS GC	10K	-	91.22
5K PDMS GC	5K	-	84.73
1K PDMS GC	1K	-	51.08
750 PEG GC	-	750	45.48



Scheme 3.1. Synthesis of functional GC resins PDMS glycidyl carbamate resin (a) and PEG glycidyl carbamate resin (b)

3.2.3. Solids content, epoxy equivalent weight, and mechanical measurements

The percentage solids of the above resins were determined according to ASTM 2369, as described in Chapter 2.2.4. The equivalent epoxy weight of the above resins was titrated following ASTM D1652, as described in Chapter 2.2.5. Mechanical measurements such as pencil hardness, König Pendulum Hardness, MEK double rubs, impact, conical mandrel tests, and tap pull-off adhesion were carried out under standards ASTM D3363, ASTM D4366, ASTM D5402, ASTM D2794, ASTM D522, and ASTM D3359, respectively.

3.2.4. Fourier transform infrared spectroscopy

A Thermo Scientific Nicolet 8700 FTIR was used to run Fourier transform infrared (FTIR) spectroscopy of GC resin and GC prepolymers. A blank potassium bromide (KBr) plate was scanned first to get the background spectrum. The GC resin and GC prepolymers solution in acetone was spread as a thin layer on the same potassium bromide (KBr) plate to collect the spectrum with the background subtracted.

3.2.5. Coating formulations

To address the critical hydrophilic-lipophilic ratio between PDMS and PEG, a wide range of molar ratios between PDMS with PEG were examined from 100:1 to 0.5:1. This work focuses on reducing the content of PEG in the coating system to reduce the amphiphilicity of the coating and improve its hydrophobicity because previous studies have shown excessive amphiphilicity. The amphiphilic functional resin was synthesized as above. Each active resin molecule had two epoxy groups used as epoxy in the system. The molar ratio of PDMS to PEG was controlled by adjusting the PDMS GC and PEG GC content. Any molar ratio of PDMS and PEG can be easily achieved. In this work, four molar ratios between PDMS and PEG were investigated. They were 100:1, 10:1, 2:1, 1:1, and 0.5:1 while keeping the weight content of PDMS in the coating between

5% and 35%. The maximum PDMS content in the coating depends on the molecular weight of the PDMS. The higher the molecular weight of PDMS used, the higher the maximum PDMS content in the coating.

The formulations of this work were divided into two parts. The first part fixes the molar ratio of PDMS and PEG to 1:1 and investigates the maximum PDMS content in each system. For example, for a system using a 10K PDMS GC, if the content of PDMS in the coating exceeded 30%, the film would not cure completely because it reduced the overall crosslink density of the coating. The mechanical properties were measured, and then the second part was that the PDMS content in the formula with the best mechanical properties was fixed, and the molar ratio of PDMS and PEG in the formulation was changed, as mentioned above. A total of 32 formulations labeled from L1 to L32 were evaluated, as Table 3.2 shows, and only 25 formulations were worthwhile to proceed to the next step. Priamine_1075 was used for all formulations as the curing agent.

Table 3.2. Formulations with a fixed weight percentage of PDMS and the variable molar ratio of PDMS and PEG.

Formulation	Formulation Type	PDMS MW ($\bar{M}_n$)	PDMS Wt.%	PEG MW ($\bar{M}_n$)	The molar ratio of PDMS to PEG	Curing agent
L1	35%-10K-1	10,000	35	750	1	Priami ne_107 5
L2	30%-10K-1	10,000	30	750	1	
L3	25%-10K-1	10,000	25	750	1	
L4	15%-10K-1	10,000	15	750	1	
L5	5%-10K-1	10,000	15	750	1	
L6	25%-10K-0.5	10,000	25	750	0.5	
L7	25%-10K-2	10,000	25	750	2	
L8	25%-10K-10	10,000	25	750	10	
L9	25%-10K-100	10,000	25	750	100	
L10	32%-10K-2	10,000	32	750	100	
L12	35%-5K-1	5,000	35	750	1	Priami ne_107 5
L13	25%-5K-1	5,000	25	750	1	
L14	20%-5K-1	5,000	20	750	1	
L15	15%-5K-1	5,000	15	750	1	
L16	5%-5K-1	5,000	5	750	1	
L17	20%-5K-0.5	5,000	20	750	0.5	
L18	20%-5K-2	5,000	20	750	2	
L19	20%-5K-10	5,000	20	750	10	
L20	20%-5K-100	5,000	20	750	100	
L21	25%-5K-100	5,000	25	750	100	
L22	37%-1K-1	1,000	37	750	1	Priami ne_107 5
L23	35%-1K-1	1,000	35	750	1	
L24	30%-1K-1	1,000	30	750	1	
L25	25%-1K-1	1,000	25	750	1	
L26	15%-1K-1	1,000	15	750	1	
L27	10%-1K-1	1,000	10	750	1	
L28	5%-1K-1	1,000	5	750	1	
L29	15%-1K-0.5	1,000	15	750	0.5	
L30	15%-1K-2	1,000	15	750	2	
L31	15%-1K-10	1,000	15	750	10	
L32	15%-1K-100	1,000	15	750	100	

3.2.6. Surface characterization

3.2.6.1. Contact Angle Characterization

Water and diiodomethane contact angles were measured by using a Kruss® DSA 100 (Drop Shape Analyzer) to determine the surface wettability and the surface energy of the coatings. The potential surface interaction with the water droplet over time was studied by measuring the static contact angle over 6 minutes, where a 2- μ L water droplet was monitored on a coating surface. Also, a tilting method was employed to study the slip angle. Each surface's advancing and receding water contact angles were measured by tilting the stage at 10°/min, where a 25- μ L water droplet was monitored on the surface. All contact angle and surface energy results for each coating were auto-calculated using Kruss® Advance software calculated following the Owens-Wendt method.

3.2.6.2. XPS

The surface elemental composition of each coating was examined by A Thermo Scientific™ K-Alpha™ X-ray photoelectron spectroscopy (XPS) instrument, which was equipped with a monochromatic Al K α (1486.68 eV) X-ray source and an Ar⁺ ion source (up to 4000 eV). A cleaned sample with size 2 mm x 2 mm was sputtered with a large Ar⁺ ion cluster with a power of 4000 eV using the MAGCIS® cluster gun before analysis for each coating. Spectra were collected at an angle normal to the surface (90°) of a 400- μ m area at a maintained below 1.5 x 10⁻⁷ Torr chamber pressure ambient temperature. The instrument's software quantified atomic concentrations of C1s, N1s, O1s, and Si2p. All atomic force microscopy images were generated using a Jupiter XR atomic force microscope (Asylum Research, Oxford Instruments, Santa Barbara, CA, U.S.A.) in Blue Drive air tapping mode using an AC240-TSA-R3 probe. Images of heights and phases were determined. Collecting images on a sample area of 100 μ m x 100 μ m of the surface of all

coatings were scanned (scan rate: 0.7 Hz; point & lines: 512; setpoint: 1.2v; integral gain:60; drive amplitude: 12.63mW; drive frequency: 63.48kHz; tip material: silicon;).

3.2.7. Biological laboratory assays

3.2.7.1. *Bacterial (Cellulophaga lytica) biofilm adhesion*

Bacterial analysis was carried out after 28 days of water immersion preconditioning. Leachate toxicity was assessed by introducing the bacterium into overnight extracts (artificial seawater with nutrients) of each coating and evaluating growth after 24 hours via a crystal violet colorimetric assay. Growth in coating leachates was reported as an absorbance ratio (600 nm) to growth control. The analysis also included a series of negative growth controls (medium + bacteria + triclosan). *C. lytica* biofilm retention analysis was assessed by a crystal violet colorimetric assay. A 5% suspension of *C. lytica* in ASW + nutrients ($\sim 10^7$ cells.ml⁻¹) were prepared, and 1 mL was added to each well of the coating plate. Plates were incubated statically at 28 °C for 24 hours to facilitate bacterial attachment and colonization. Plates were rinsed three times with DI water and stained with crystal violet. Images were taken after staining, then the crystal violet was extracted in 33% acetic acid (AA), and the resulting eluates were measured for absorbance at 600nm. Biofilm retention was reported as the mean absorbance value of three replicate samples. Error bars represent one standard deviation of the mean. Water jet adhesion was carried out after 24 hours of bacterial biofilm growth. The first column of each plate was not treated and served as the measure of biofilm growth before water jetting. The second and third column of each coating was jetted for 5 seconds at a pressure of 10 psi and 20 psi, respectively. Biofilm adhesion was reported as a function of biomass remaining on the material surface after treatment with each pressure indicated above. Error bars represent one standard deviation of the mean. ²⁻³

3.2.7.2. Growth and release of fouling microalgae, *Navicula incerta*

All microalgae analysis was carried out after 28 days of water immersion. The growth of *Navicula incerta* was assessed by introducing algae into overnight extracts (artificial seawater with nutrients) of treatment and evaluating growth after 48 hours via fluorescence of chlorophyll. Growth in coating leachates was reported as a fluorescence ratio to a growth control (fresh nutrient medium). A negative growth control (medium + bacteria + triclosan) was also included in the analysis. The initial cell attachment of algae was assessed before water jet adhesion analysis. Algae were diluted to an OD of 0.03 at 660 nm in artificial seawater (ASW) supplemented with nutrients. 1 mL was added to each well of the plate and allowed to incubate statically for 2 hours to facilitate cell attachment. Algal cell attachment was quantified by fluorescence measurement of DMSO extracts of chlorophyll. Cell attachment was reported as fluorescence intensity (relative fluorescence units). Error bars represent one standard deviation of the mean. Water jet adhesion was carried out after 2hr initial cell attachments. The first column of each plate was not treated and served as the measure of cell attachment after 2 hr. The second and third column of each coating was jetted for 10 seconds at a pressure of 10 psi and 20 psi, respectively. Algal adhesion was reported as a function of biomass remaining on the material surface after treatment with each pressure indicated above. Error bars represent one standard deviation of the mean.³⁻⁴

3.2.7.3. Adult barnacle 2-week reattachment and adhesion (*A. amphitrite*)

Coated panels (4" x 8" aluminum) were preconditioned/aged for 28 days via water immersion before the barnacle attachment assessments were conducted. Then six barnacles were immobilized to the surface of the coatings. Coatings were analyzed after two weeks of attachment with daily brine shrimp feedings. A hand-held digital force gauge, mounted to an automated stage, was used to measure the peak force of release for each barnacle. The number of barnacles that

could attach to each surface was recorded, and the attachment efficiency was calculated (percent). Adhesion strength in shear was calculated by dividing the measured force required to remove the barnacle by the basal area and reported in Megapascals (MPa). Each data point is the mean value of the total number of barnacles that are attached to the coating surface. Error bars represent one standard deviation of the mean.⁵⁻⁶

3.3. Results and Discussion

Chapter 2 shows amphiphilic coatings have good antifouling (AF) and fouling release (FR) performance and robust surfaces. The work in Chapter 2 extended the previous work in our team. More curing agents were used to cross-link with the new novel isocyanate-free coating system, which still contains urethane linkages to study their effects. In Chapter 2, PDMS and PEG continue to provide the amphiphilic domains, and the weight ratio of PDMS and PEG was fixed at 1:1 as in the previous work. Because the molecular weight of PDMS used in the earlier work was much larger than that of PEG, the actual molar ratio of PDMS and PEG was small. For example, for the coatings that used 10K $\bar{M}_n$ PDMS and 550 $\bar{M}_n$ PEG with a 1:1 weight ratio, the molar ratio of PDMS and PEG was 0.055:1, which leads to excess anti-fouling and fouling release performance of the surface for some marine organisms. Biological measurement results in Chapter 2 showed that most of the coatings with PDMS and PEG at a 1:1 weight ratio had very good, even some “excess” anti-fouling and fouling release performance against *C.lytica*. For example, the coatings containing 10% 5K 550 were almost immune to *C.lytica*. However, the ocean is complex, and marine organisms have a variety of strategies on the surface. This work attempted to adjust the balance of the anti-fouling and fouling release performance against different marine organisms by adjusting the molar ratio of hydrophobic and hydrophilic

components in the coatings. We tried to find the critical hydrophilic-lipophilic ratio (CHLR) between the amphiphilic domains.

The glycidyl-carbamate (GC) resin containing epoxy groups, the PDMS GC resins, and PEG GC resin containing the hydrophobic and hydrophilic domains were prepared similarly by the reaction between IPDI trimer, glycidol, hydroxyl-terminated PDMS and hydroxyl-terminated PEG respectively. Three molecular weights of PDMS ($1,000 \bar{M}_n$, $5,000 \bar{M}_n$, and $10,000 \bar{M}_n$) were used to synthesize the PDMS GC resins, and PEG ($750 \bar{M}_n$) was used to synthesize the PDMS GC resin. A total of three PDMS GC resins and one PEG GC resin were synthesized. The molar ratio of the coating hydrophilic (PEG) and hydrophobic (PDMS) was adjusted by adjusting the PDMS GC and PEG GC ratio in the formulations. Priamine_1075 was used as the curing agent, and the effect of the molecular weight of PDMS used in the coatings, the weight content of PDMS in the solid coating, and the molar ratio of PDMS and PEG on performance were investigated. The content of PDMS was first screened with a fixed molar ratio at 1:1 of PDMS and PEG. The higher the content of PDMS, the poorer the mechanical properties of the coating would be. Coatings containing PDMS with different molecular weights have a corresponding maximum PDMS content. According to this maximum content, the corresponding adjusting of the PDMS and PEG molar ratio was examined. Some formulations were selected for surface characterization and laboratory biological assessments.

The success of the synthesis of GC resin, PDMS GC resins, and PEG GC resin was confirmed by the complete reaction of the isocyanate-functional IPDI trimer with the hydroxyl-terminated compounds, which was indicated in the FTIR spectra (Figure 3.1). The absence of the signature isocyanate peak at 2250 cm^{-1} and the appearance of the secondary amine (due to the formation of the urethane linkage) and the peaks at 3350 cm^{-1} and 1128 cm^{-1} respectively, of the

GC resin and PDMS GC resins, and PEG GC resin indicated the completion of the reaction. The stretching for PDMS at 1030 cm^{-1} and PEG at 1105 cm^{-1} in the PDMS GC resins and PEG GC resin spectrum showed the amphiphilic chains, respectively.

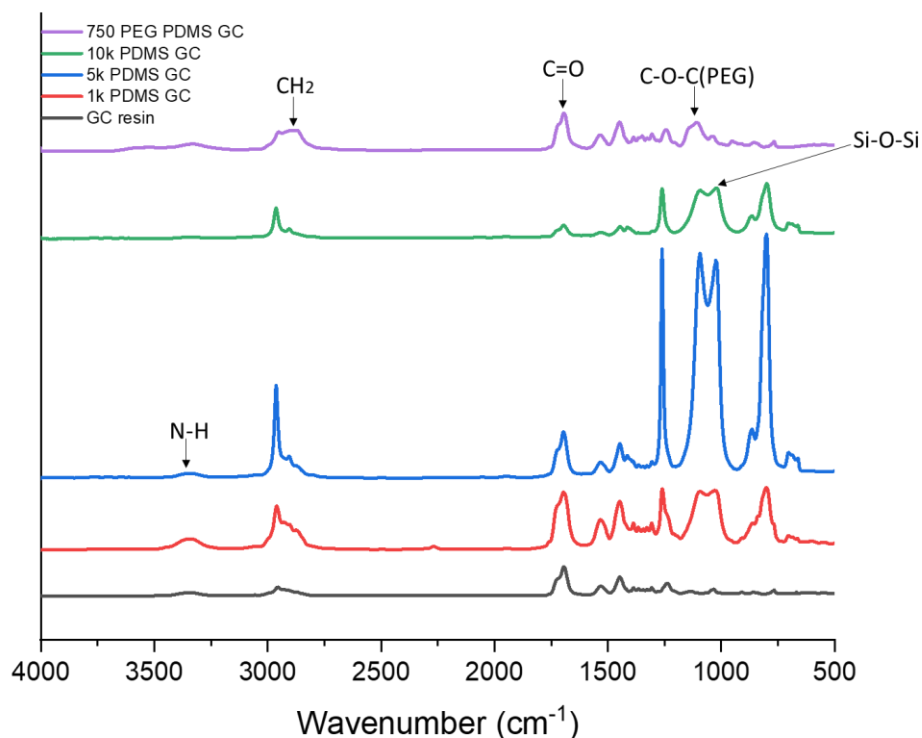


Figure 3.1. FTIR Spectrum for resins: GC resin, 1K PDMS GC, 5K PDMS GC PDMS, 10K PDMS GC, and 750 PEG GC.

The epoxy equivalent weight (EEW) values of GC resin, PDMS GC resins, and PEG GC resin were titrated ASTM D1652, and the results are shown in Table 3.2. The resins had almost the same EEW as the theoretical value, as expected. The difference (about 10%) between the titrated EEW and the theoretical value may be caused by the measurement error. The EEW of GC resin was much lower than the amphiphilic-modified GC resins, was because of the much higher amount of glycidol used in the synthesis of GC resin than the synthesis of amphiphilic-modified GC resins, suggesting the availability of more epoxy functional groups in GC resin and

limited presence of epoxy functional groups in the amphiphilic-modified GC resins, which resulted in the little concentration of amphiphilic domain in the solid coating system.

Table 3.3. Epoxy equivalent weight results.

Materials	Theoretical EEW. g/eq	Titrated EEW, g/eq
GC resin	328.38	361.5
10K PDMS GC	5484.20	5028.17
5K PDMS GC	2953.60	2772.04
1K PDMS GC	978.87	906.44
750 PEG GC	824.45	765.55

Most formulations, except those with very high PDMS content, had excellent mechanical properties. Overall, the coatings with high content of PDMS had poor mechanical properties or could not form a smooth film. This work prepared three PDMS GC resins with three different molecular weight PDMS (1K, 5K, 10K) as described in the formulation design above. In this work, the molar ratio of PDMS and PEG was fixed at 1:1 first to investigate the effect of PDMS content. The increase of PDMS content would reduce the crosslinking density of the entire formulation because PDMS GC resin had little reactivity group – the epoxy group. Those three PDMS GC resins had their own critical PDMS content. For the PDMS GC resin using 10K PDMS, when the content of PDMS exceeds 30%, no film could be formed because the crosslinking density of the coating system was too low and the poor compatibility with curing agents. When using PDMS GC resin synthesis with 5K PDMS, if the content of PDMS exceeds 25%, no film could be formed. But it was worth noting that a smooth film was formed using PDMS GC resin with 1K PDMS. The content of PDMS even exceeds 37%, perhaps because the equivalent epoxy weight of 1K PDMS resin itself is sufficient to create a smooth film. There was not much change in the coatings' mechanical properties when the content of PDMS was fixed, and the molar ratio of PDMS and PEG was adjusted. That was because the equivalent epoxy

weight of PEG GC resin was not very high-765.55, and the proportion of its dosage in the entire coating formulation was small, so the effect on the crosslinking density of the coatings was limited.

Table 3.4. Mechanical properties

Formulation	Pencil Hardness	Impact Test		MEK Double Rubs	Conical Mandrel	König Pendulum Hardness	Crosshatch Adhesion
		Forward	Reverse				
L1	No film formation						
L2	No film formation						
L3	2H	28	7	130	Fail	95	3B
L4	H	22	4	140	Fail	100.7	2B
L5	5H	15	4	87	Fail	99.7	0B
L6	H	28	3	30	Pass	38	5B
L7	2H	18	2	35	Pass	46	3B
L8	2H	18	5	90	Pass	42	3B
L9	2H	20	2	90	Fail	13	3B
L10	No film formation						
L12	No film formation						
L13	No film formation						
L14	5H	15	3	280	Fail	33.33	4B
L15	2H	11	1	>400	Fail	109.3	0B
L16	6H	16	3	241	Fail	64	0B
L17	2H	14	Fail	59	Fail	36	5B
L18	3H	28	11	170	Fail	35	2B
L19	5H	26	10	130	Fail	48	0B
L20	5H	29	3	130	Fail	53	0B
L21	2H	21	8	50	Fail	46	4B
L22	3B	6	Fail	3	Pass	39	5B
L23	B	19	Fail	7	Pass	46	5B
L24	F	>43	>43	11	Pass	75	5B
L25	F	23	3	25	Pass	82	5B
L26	H	13	Fail	30	Pass	120	5B
L27	4H	28	17	120	Fail	142	3B
L28	5H	13	1	>400	Fail	144	0B
L29	B			20	Pass	93	5B
L30	3H	7	Fail	100	Fail	116	0B
L31	3H	12	8	200	Fail	135	0B
L32	2H	2	Fail	148	Fail	113	0B

Static contact angle measurements of water and diiodomethane were done at zero time and after 6 minutes to monitor changes due to the change over time. Surface energy for each surface was calculated using the Owens-Wendt method. Most of the water contact angle of coatings at the beginning was higher than 100 degrees, suggesting the hydrophobic character of the surface. The initial water contact angles of the coatings using 10K PDMS, 5K PDMS, and 1K PDMS resin showed a decreasing trend, although the trend was not so evident. A clear trend could be observed that at the same molar ratio of PDMS and PEG, the higher the content of PDMS in the coating, the larger the water contact angle, and the water contact angle can exceed 110 when the content of PDMS in the coating exceeds 25%. And when the content of PDMS in the coating was fixed, the larger the molar ratio of PDMS to PEG, the larger the water contact angle. For the coating using 10K PDMS GC resin, when the molar ratio of PDMS and PEG was fixed at 1:1, as the content of PDMS in the coating decreased, the initial water contact angle had an upward trend. In contrast, the water contact angle after 6 minutes showed a downward trend. The diiodomethane contact angle also had the same trend, but the trend was not obvious. The initial surface energy showed a decreasing trend, but the surface energy after 6 minutes showed an increasing trend. For the coating using 5K PDMS GC resin, when the molar ratio of PDMS and PEG was fixed at 1:1, as the content of PDMS in the coating decreased, the initial water contact angle and the water contact angle after 6 minutes both showed a slight downward trend. The diiodomethane contact angle value was relatively stable. The initial and surface energy after 6 minutes was also stable. For the coating using 1K PDMS GC resin, when the molar ratio of PDMS and PEG was fixed at 1:1, as the content of PDMS in the coating decreased, the initial value of the water contacts angle and the water contact angle after 6 minutes was relatively stable. However, the initial diiodomethane contact angle and the value after 6 minutes both showed an upward trend. The initial and surface

energy after 6 minutes were mostly stable. When the content of PDMS was fixed, the water contact angle, diiodomethane contact angle, and surface energy did not change much, which may be related to the level of the investigated factors.

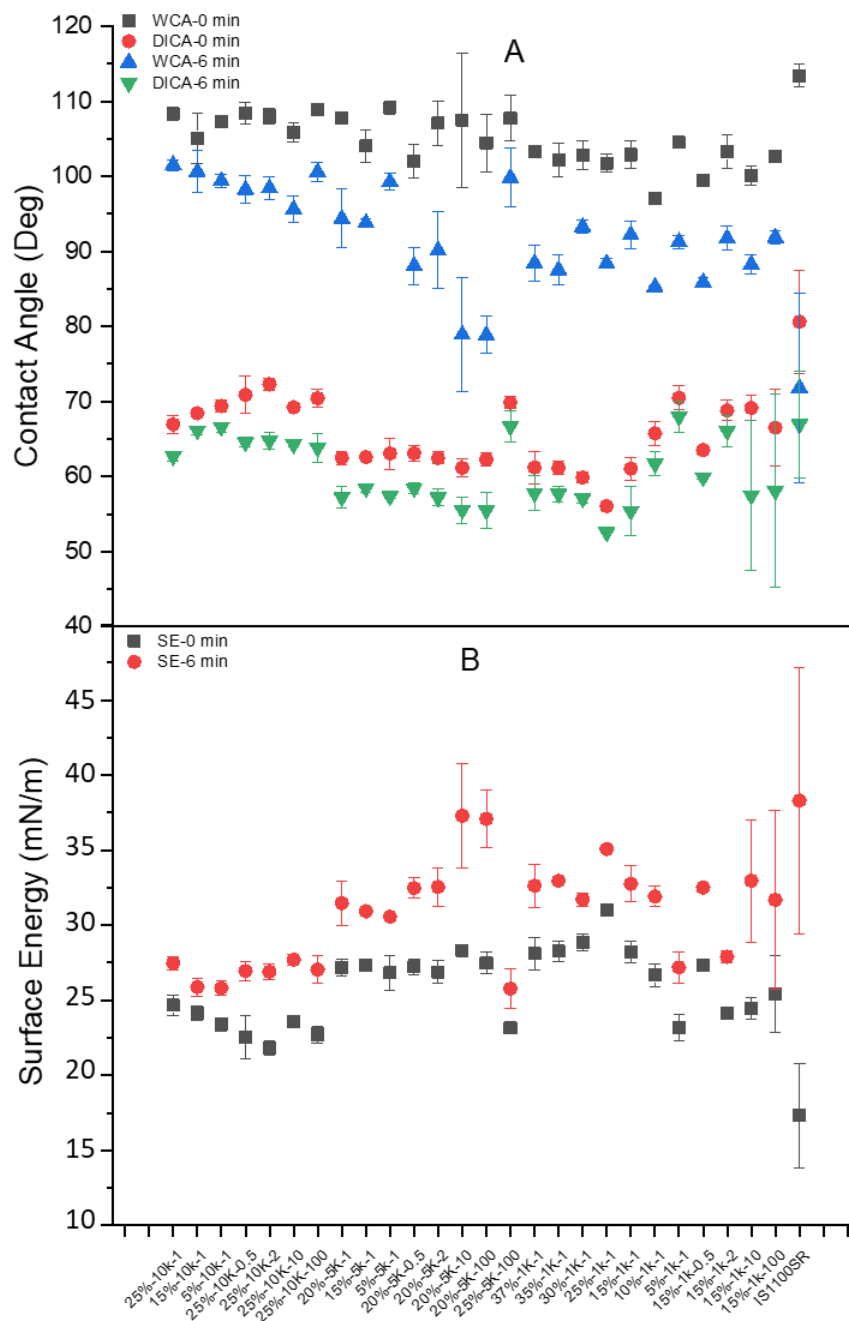


Figure 3.2. Static contact angle data for coatings: Water contact angles (WCA) and methylene iodide contact angles (MICA) at 0 minute and 6 minutes and their surface energy, the x-axis is labeled to specify the formulations and its components, including PEG MW, PDMS MW, wt.% of PDMS, and the molar ratio between PDMS and PEG. The error bars are standard deviations of three measurements.

XPS was performed on the coatings used to observe the atomic content on and just below the surface, indicating amphiphilicity. Generally, the Si atom content on the surface of coatings containing 10K PDMS was higher than those with 5K PDMS and much higher than those with 1K PDMS. For coatings containing PDMS with the same molecular weight, when the molar ratio of PDMS and PEG was fixed, the higher the content of PDMS, the higher the silicon content on the coating surface. This trend weakens as the molecular weight of PDMS increases. For coatings containing 10K PDMS and 5K PDMS, silicon content on the surface decreases when the molar ratio of PDMS and PEG increases. The higher the molecular weight of PDMS, the greater the effect of the molar ratio of PDMS and PEG. For example, for the coatings containing 10K PDMS, when the molar ratio of PDMS and PEG was 1:1, the surface silicon content of the coating containing 25% PDMS was higher than that of a coating containing 15% PDMS, while PDMS and PEG were 0.075:1 for the coating containing 15% PDMS, which on the contrary, the content of silicon on the surface was higher than that of the coating containing 25% PDMS with the molar ratio of PDMS and PEG was 1:1. This was attributed to the increased repulsion to PDMS when the PEG content in the system increases, making it easier for PDMS to reach the surface. But for the coatings containing 1K PDMS, the molar ratio of PDMS and PEG increases, and the content of silicon on the surface of the coating increases instead, probably because 1K PDMS was very compatible with the system. PEG has no repulsion to PDMS or was not noticeable. The content of the system PDMS mainly determined the silicon content on the surface of the coating.

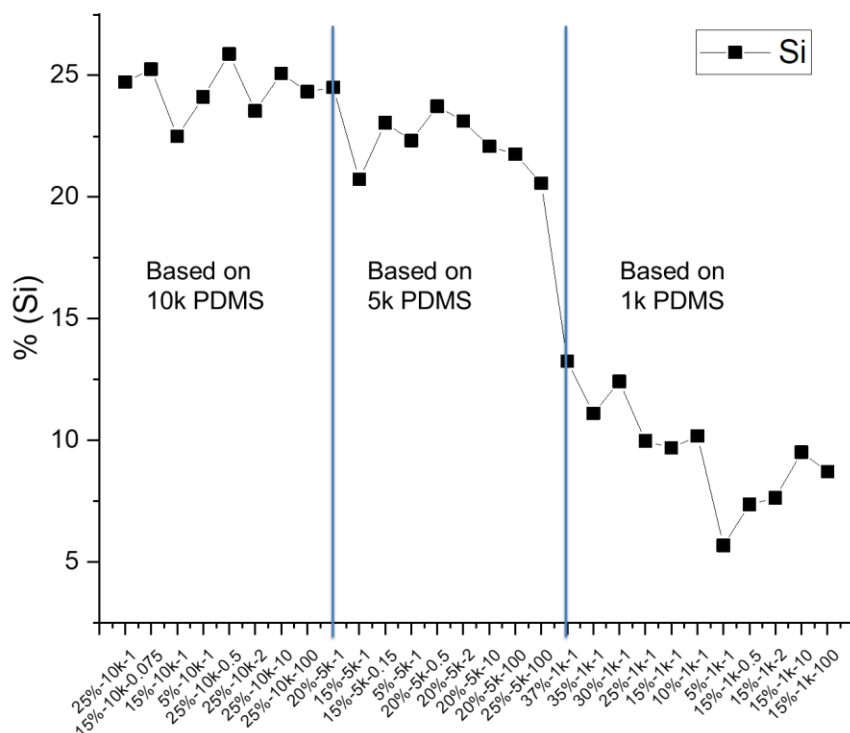


Figure 3.3. XPS data: Si atomic content on and just below the surface, indicating the effect of molecular weight of PDMS and PEG on surface self-assembly of coatings. The effect of curing agents on the silicon atom content on the surface suggested that the polarity of curing agents played a significant role in self-stratification and surface composition.

The AFM micrographs in Fig. 3.4.-Fig. 3.6. show the surface morphologies of amphiphilic domains of the set of coatings using 1K, 5K, 10K PDMS GC, and 750 PEG GC resins as a hydrophobic functional material. As discussed in Chapter 2, smaller phase angles indicated harder (PEG), and higher phase angles indicated the presence of softer materials (PDMS). The top image is the Height image, and the bottom image is the Phase image. For the coatings using 10K PDMS GC resin, when the molar ratio of PDMS and PEG was fixed at 1:1, the PDMS domain became smaller and more uniform as the PDMS content decreased. When the content of PDMS was 5%, the PDMS domain was evenly distributed on the surface, and the size of PDMS domains was tiny and had a small range. For the formula using 5K PDMS GC resin,

when the molar ratio of PDMS and PEG was at 1:1 and when the content of PDMS was 20%, the size and distribution of the PDMS domain were very uniform. When the content of PDMS was fixed and adjusting the molar ratio of PDMS and PEG was, as the molar ratio increased, the changing trend of the PDMS domain in the formulations using 10K PDMS GC resin was not so obvious, which may be attributed to the dominant role of PDMS with a molecular weight of 10K. The changing trend of the PDMS domain in the coatings using 5K PDMS GC resin was apparent. As the molar ratio increased, the size and distribution of PDMS were more uniform. The morphology of the domain changed significantly if the molar ratio of PDMS and PEG was at the exact content of PDMS at 20%. The domain size was the most uniform when the molar ratio of PDMS and PEG was 10:1. This was due to the reduction of PEG content, which reduced the force of pushing PDMS together. It was worth noting that in the coatings using 1K PDMS GC resin as the hydrophobicity provider, no matter whether it was changing the content of PDMS or changing the molar ratio of PDMS and PEG, the amphiphilic domain was not obvious, the boundary between the hydrophobic domain (PDMS) and the hydrophilic domain (PEG) did not seem to be so precise, probably because the compatibility of 1K PDMS resin with the system was very well, which was confirmed by that both the coatings and film were utterly transparent.

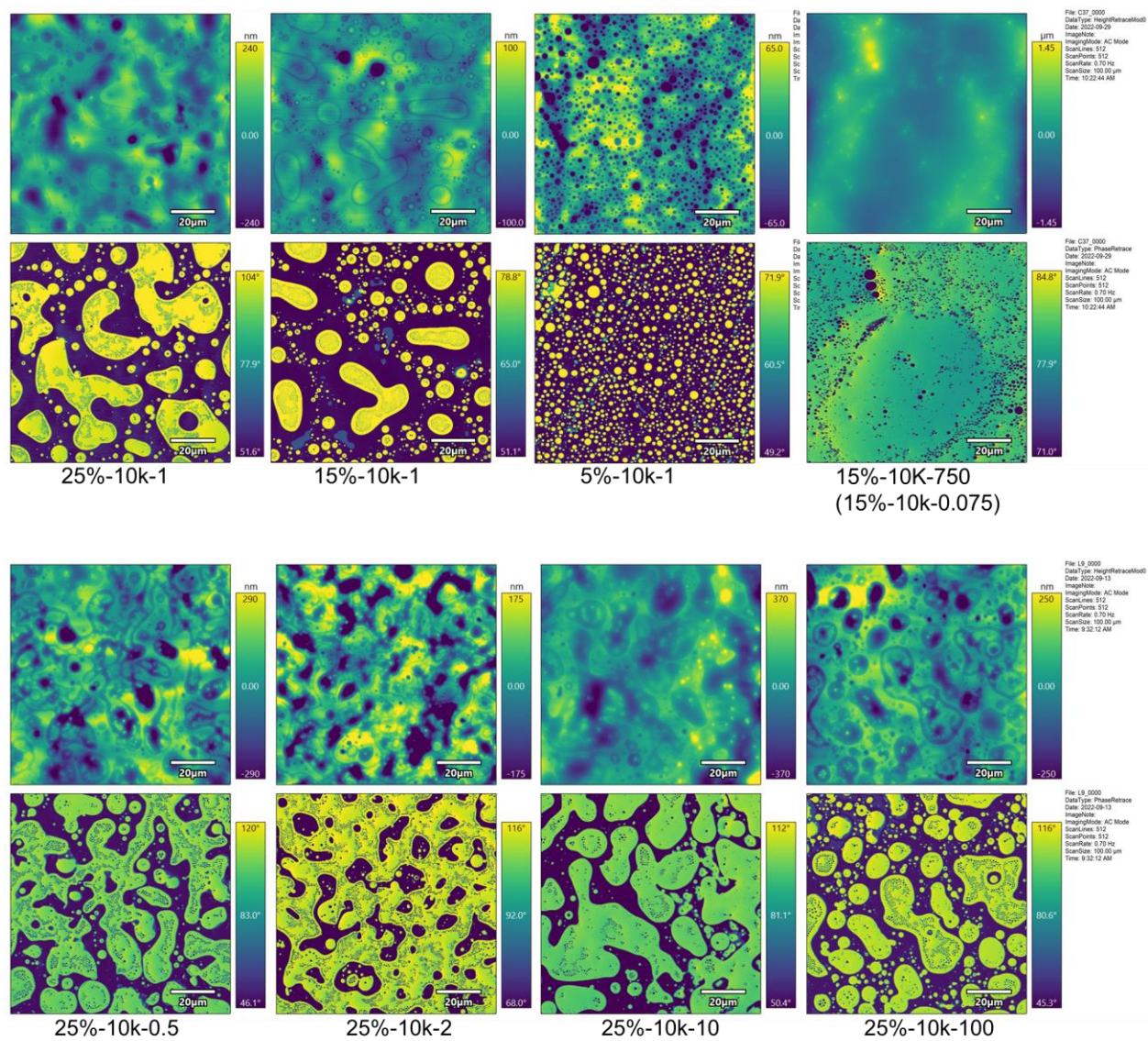


Figure 3.4. AFM Height and Phase images of coatings contain 10K PDMS over 100 x 100 μm.

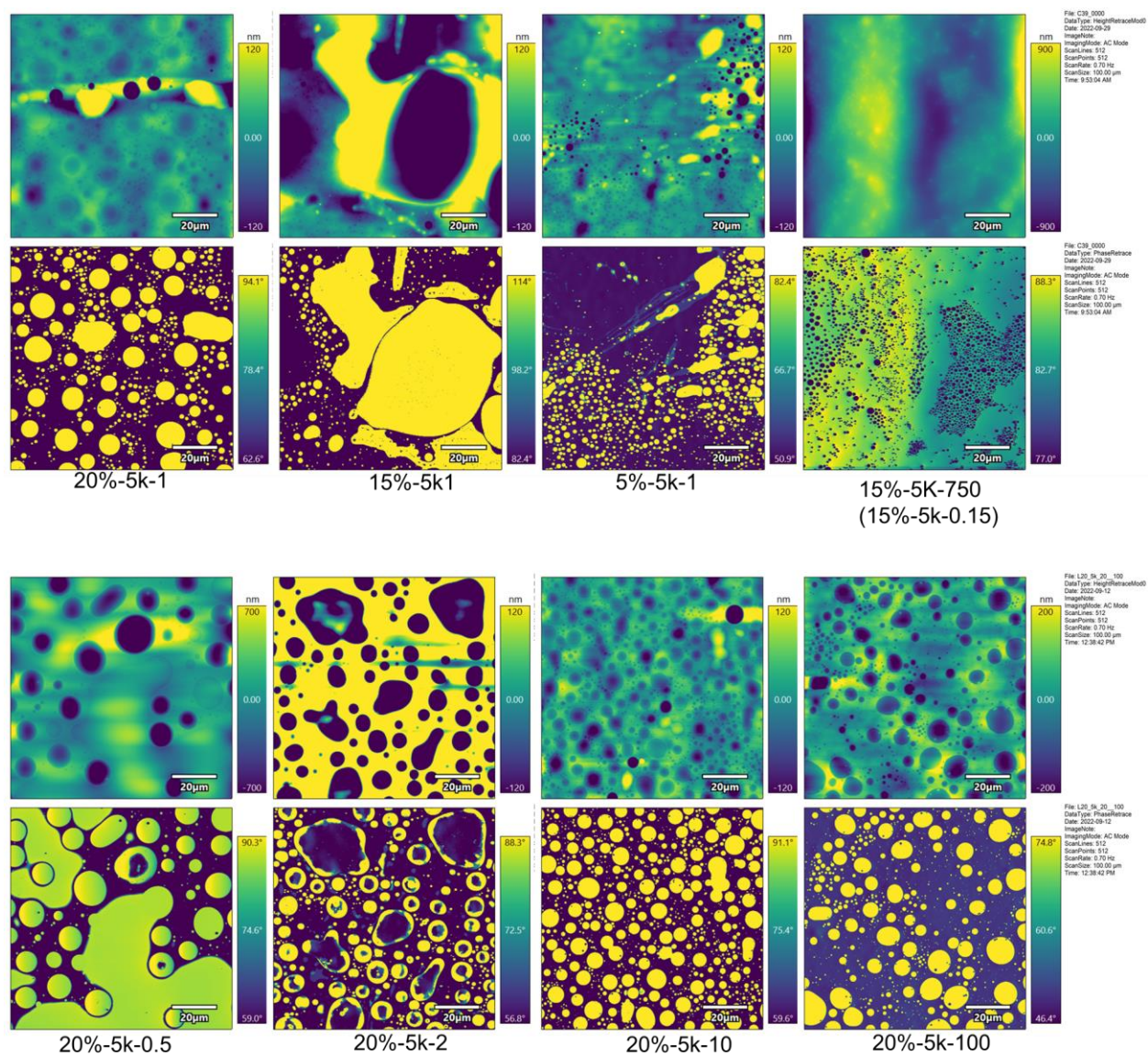


Figure 3.5. AFM Height and Phase images of coatings contain 5K PDMS over 100 x 100 μm.

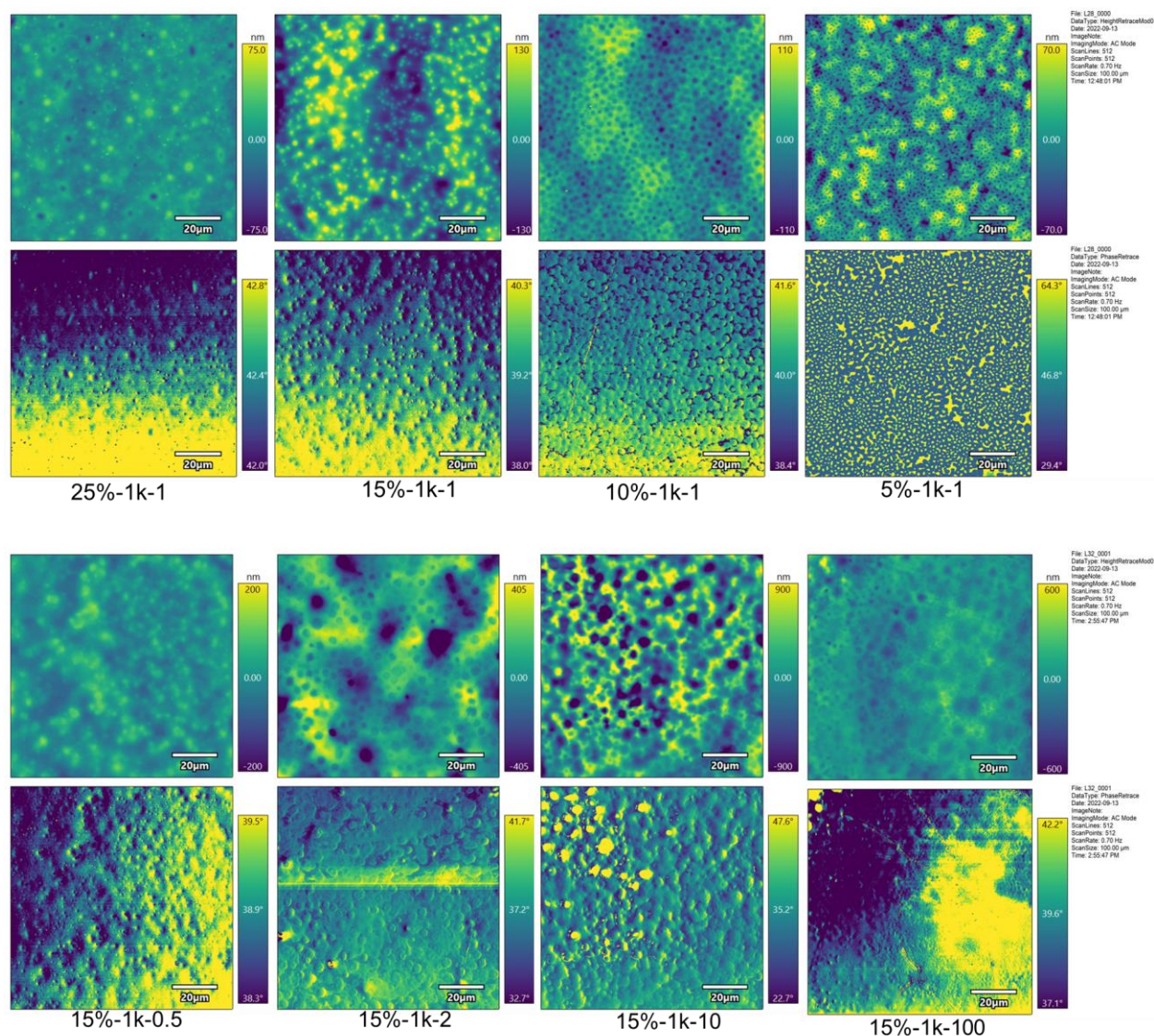


Figure 3.6. AFM Height and Phase images of coatings contain 1K PDMS over 100 x 100 μm.

Diatom *Navicula incerta* and bacterium *C. lytica* were used to evaluate all coatings' laboratory fouling release performance. Coatings were immersed in water for 28 days for leachate toxicity assessment to make sure the coatings measured fouling release performance without any toxic ingredients as the same procedure as described in Chapter 2. The cell attachment for *N. incerta* and *C. lytica* was stimulated by static incubation for two hours under ambient conditions in the dark, and coating surfaces were then subjected to water-jet treatments at 10 and 20 psi water pressure for ten seconds. The relative fluorescence (RFU) measured from

the extracts was directly proportional to the biomass remaining on the coating surfaces after water-jetting, which was used to evaluate the *N. incerta* and *C. lytica* fouling release performance.

Figure 3.7. shows that all formulations' initial extent of diatom biofouling was less than that of all controls, indicating better anti-fouling performance. For the coating using 10K PDMS, when the molar ratio of PDMS and PEG was fixed at 1:1, the formulation with 15% PDMS had the best fouling release performance. When the content of PDMS was set at 25%, and the molar ratio of PDMS and PEG was 1:1, the coating had the best fouling release performance of all coatings based on 10K PDMS GC resin. For the coatings using 50K PDMS, when the molar ratio of PDMS and PEG was fixed at 1:1, the fouling release performance of the coating became better as the PDMS content decreased. When the content of PDMS was set at 20%, and the molar ratio of PDMS and PEG was 2:1, the coating had the best fouling release performance of all coatings based on 5K PDMS GC resin both at 10 psi and 20 psi water pressure levels. It was worth noting that for coatings using 1K PDMS GC resin when the molar ratio of PDMS and PEG was fixed at 1:1, the formulations with PDMS content exceeding 5% had comparable fouling release performance to IS1100SR, where the coating with 25 wt.% PDMS content had the best fouling release performance and was even better than IS1100SR. However, when the content of PDMS was fixed at 15%, and the molar ratio of PDMS and PEG exceeded 1:1, the coating's fouling release performance deteriorated. Overall, regardless of the molecular weight of PDMS and the content in the coating, when the molar ratio of PDMS and PEG was 1:2, the coatings had the best diatom biofouling release performance.

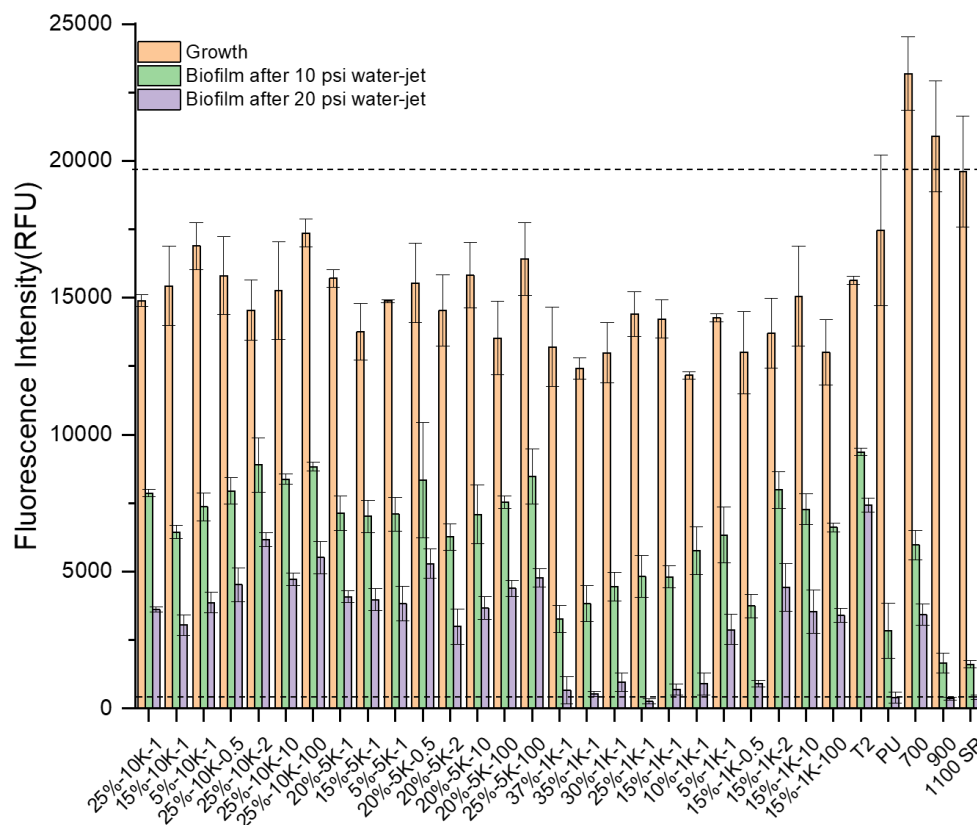


Figure 3.7. *N. incerta* data for biofilm growth (yellow bar), release at 10 psi (green bar) and release at 20 psi (purple bar). The X-axis is labeled to specify the formulations and their components, such as wt.% of PEG and PDMS and controls. The error bars are standard deviations of six measurements.

As discussed in Chapter 2, *C. lytica* is a known biofouling microorganism with an affinity for a wide range of hydrophilic and hydrophobic surfaces. The coatings in Chapter 2 had very good amphiphilicity and excellent fouling release performance compared to the controls. The fouling release performance at both 10 and 20 psi water pressure levels of most of the coatings in this work was much better than the controls T2, PU, Intersleek 700, and Intersleek 900, and similar or better than Intersleek 1100SR, which because that most of the coatings in this work had very good amphiphilicity. The PEG content of coatings in this work was much less than in Chapter 2, and the overall *C. lytica* fouling release performance was worse than the coatings in

Chapter 2. All formulations' initial extent of biofouling was better than all controls, except the formulation of 25%-10K-2. In particular, the formulations containing 20% PDMS with PDMS and PEG molar ratios of 0.5, 2, 10, and 100 had an initial biofilm that was less than that of IS1100SR, showing an excellent anti-fouling function. The fouling release performance was the best for the coatings using 10K PDMS GC resin when the molar ratio of fixed PMDS and PEG was 1:1 and the PDMS content was 15%. When the content of PDMS was fixed at 25%, the performance was the best when the molar ratio of PDMS to PEG was 2. For the coatings using 5K PDMS GC resin, when the molar ratio of PMDS and PEG was fixed at 1:1, the fouling release performance had little difference when the PDMS content changes, which was similar to the fouling release performance of IS1100SR. The fouling release performance was the best when the PDMS content was 20%. When the content of PDMS was fixed at 20%, when the molar ratio of PDMS to PEG was higher than 2, the molar ratio increased, and the fouling release performance deteriorated. The fouling release performance was the best when the molar ratio of PDMS to PEG was 2:1. For the coatings using 1K PDMS GC resin, the molar ratio of fixed PMDS and PEG was 1:1, and the content of PDMS was 30%, the fouling release performance was the best. When the content of PDMS was fixed at 15%, the fouling release performance was worse than that of 30%, and when the molar ratio of PDMS to PEG was 100:1, the fouling release performance was the best. It was worth noting that for coatings containing 10% PDMS, the molar ratio of PDMS and PEG was 1, with no biofilm remaining after both 10 psi and 20 psi water-jet levels.

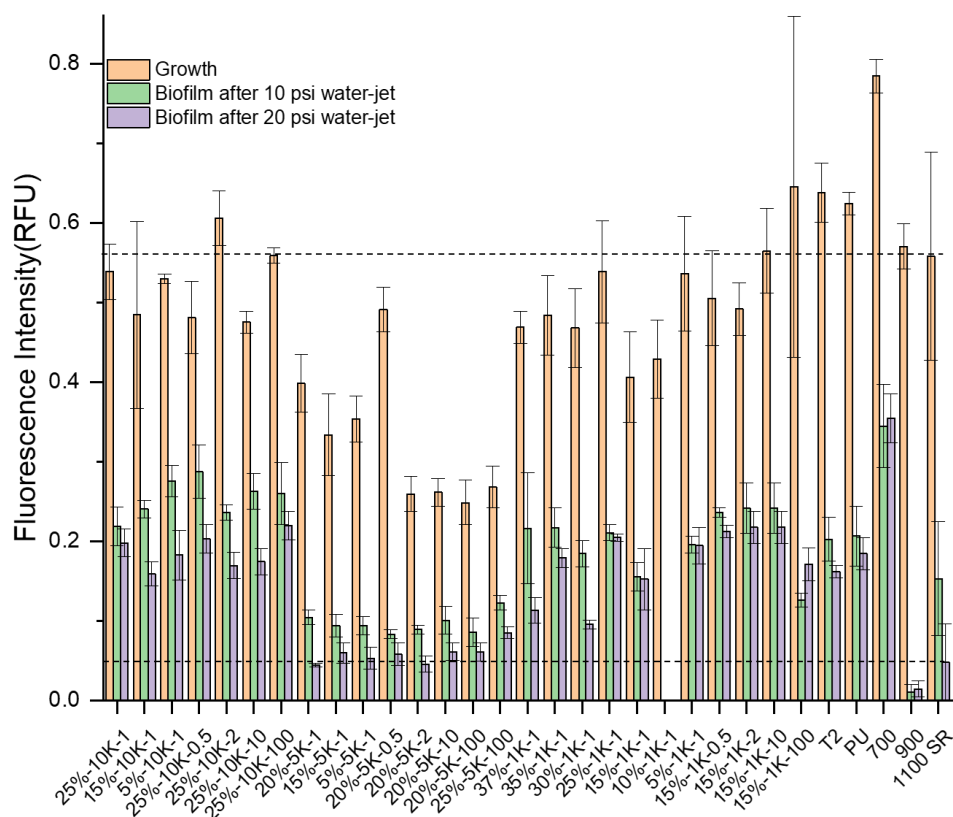


Figure 3.8. *C. lytica* data for biofilm growth (yellow bar), release at 10 psi (green bar), and release at 20 psi (Purple bar). The X-axis is labeled to specify the formulations and their components, such as wt.% of PDMS, the molar ratio of PDMS and PEG, and controls. The error bars are standard deviations of six measurements.

Four coatings with balanced good biofouling removal performance were selected for barnacle adhesion assessment. The number of barnacles reattached to the coating 5%-5K-1 and 20%-5K-2 surface were the same as the commercial coating (Fig. 3.9.). The coatings with higher molecular weight PDMS (5K $\bar{M}_n$ and 10K $\bar{M}_n$) had slightly poorer barnacle release performance than the commercial coating. The barnacle adhesion of coating 25%-1K-1 was much higher than the commercial coating. Half of them were broken during the barnacle adhesion assessment indicating strong adhesion, probably due to the low silicon content on the coating surface and the smooth surface (Fig. 3.3.). Coatings 5%-5K-1 had the best barnacle release performance among

these for selected coatings, demonstrating that the size of the hydrophobic domain probably played an important role.

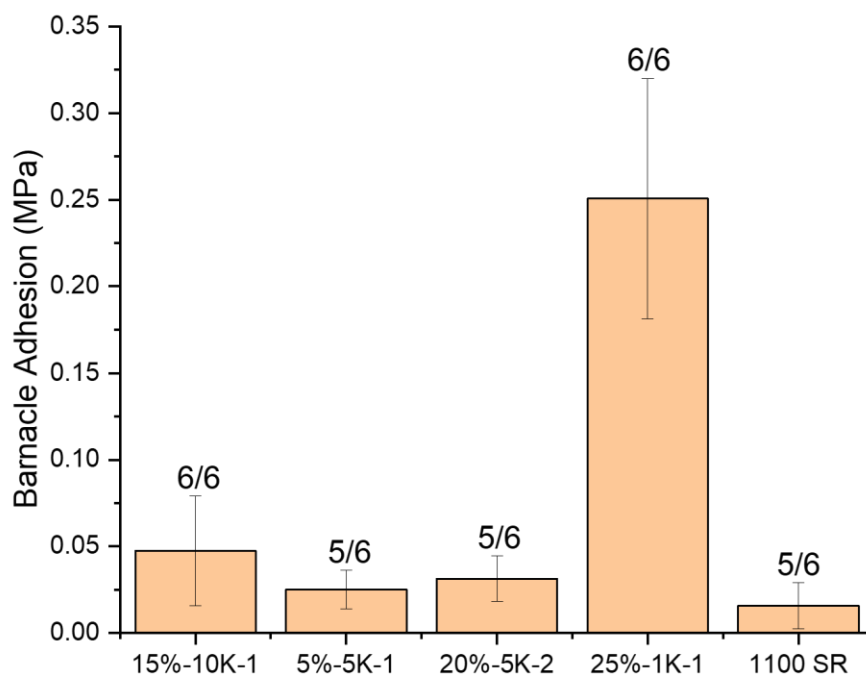


Figure 3.9. Reattached barnacle (*A. amphitrite*) adhesion strength data. Six barnacles were used for each reattachment. The number of attached barnacles out of six is labeled as a ratio for each system. Each bar shows the average adhesion strength based on the number of successfully reattached barnacles. The X-axis is labeled to specify the formulations and their components: PDMS MW, wt.% of PDMS, and the molar ratio of PDMS and PEG. The error bars are standard deviations.

3.4. Conclusions

The glycidyl-carbamate resin, PDMS glycidyl-carbamate resin, and PEG glycidyl-carbamate resin were successfully synthesized by the reaction of IPDI trimer resin with glycidol, and PDMS or PEG, converting the NCO functional group to epoxy group, which can reduce health hazards when using by works. The FTIR spectrum confirmed the completed reaction. A total of 32 coatings were prepared in this work to study the factors that affect the anti-fouling and

fouling release performance. The factors include the molecular weight of PDMS, its amount in the system, and the molar ratio of PDMS and PEG. The surface study showed that the molecular weight of PDMS and its content in the system and the molar ratio of PDMS and PEG played a significant role in the self-assembly behavior of the coatings. Contact angle measurements indicated that amphiphilic coatings have a dynamic surface. XPS results demonstrated the effects of the molecular weight of PDMS and its amount in the system and the molar ratio of PDMS and PEG on the coatings' self-stratification. AFM confirmed the presence of amphiphilic domains on the coating surface. The shape of domains depends much on the molecular of PDMS, their amount, and the molar ratio of PDMS and PEG. The biological measurements showed that the amphiphilic coating had promising anti-fouling and fouling release performance. The critical molar ratio of PDMS and PEG depends much on the PDMS molecular weight. Overall, the coatings with the PDMS and PEG molar ratio at 1:1 to 2:1 had better anti-fouling and fouling release performance balance.

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CHAPTER 4. CONCLUSIONS AND FUTURE WORK

4.1. Conclusions

The work in Chapter 2 aimed to continue previous work to design an amphiphilic glycidyl-carbamate (GC) coating system. It examined the effects of several factors, including the molecular weight of PDMS and PEG, the weight concentration of PDMS/PEG, and the curing agent type. This system maintains polyurethane's properties and is less hazardous to workers' health because it does not contain isocyanate in the final formulation, which benefits from the urethane linkage in GC resin. The epoxy group in this system provides formulation diversity. In addition, the amphipathic nature of this system, which AFM confirmed, improves FR performance against a wide range of marine organisms. The curing agent's polarity greatly influenced the content of silicon atoms on the coating surface, hence the surface's domains. Under the same conditions, a highly polar curing agent made it easier for PDMS to reach the surface. The domains on the surface also depend much on the molecular of PDMS and PEG, their amount. The biological measurements showed that the amphiphilic coatings had promising anti-fouling and fouling release performance. Especially for biofouling microorganisms with an affinity for a wide range of surfaces, both hydraulic and hydrophobic, such as *C. lytica*, the coating in this work had excellent fouling release performance, which is attributed to the high 'confused' amphiphilic surface. The work in Chapter 3 aimed to study the critical molar ratio of PDMS and PEG. The molar ratio of PDMS and PEG played a significant role in the self-assembly behavior of the coatings. The XPS results showed that the molar ratio of PDMS and PEG and the molecular weight of PDMS play an essential role in affecting the silicon content on the surface. AFM confirmed the effects. The biological measurements demonstrated that the amphiphilic coating had promising anti-fouling and fouling release performance. The critical

molar ratio of PDMS and PEG depends much on the PDMS molecular weight. Overall, the coatings with the PDMS and PEG molar ratio at 1:1 to 2:1 had better anti-fouling and fouling release performance balance.

4.2. Future Work

The results of this research indicated that the polarity of the curing agent, the molecular weight of PDMS and PEG, and the molar ratio of PDMS and PEG significantly affect the silicon content on the coating surface, which in turn affects the anti-fouling and fouling release performance of the coating. Experiments had shown that when 1K PDMS was used, the silicon content on the coating's surface changed considerably. It is possible to use a mixture of 1K PDMS GC resin with a higher molecular weight, such as 5K PDMS GC resin, to adjust the silicon content on the surface of the coating. Another advantage of using 1K PDMS GC resin is its compatibility with the system. The coating can be made into a transparent or stable state, thus eliminating the influence of the preparation process. The molar ratio of the critical ratio of PDMS and PEG is related to the selected PDMS, PEG, and their dosage. After determining the type of PDMS and PEG, optimization of their molar ratio to determine the optimal value for this system chosen can be carried out. In addition, biological experiments can be carried out on a broader range of marine organisms to investigate the modified system's wider anti-fouling and fouling release performance. The formula with excellent performance can be selected for further field tests.

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